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EXPLORATIVE ANALYSIS OF THREE DIMENSIONAL GAIT DATA FOR
DISTINCTIVE CLUSTERING IN WALKING BEHAVIOUR IN THE
FIGURE-OF-8 WALKING TEST

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1 Introduction

Roessingh Research & Development (RRD) is the scientific research department of Roessingh Center for Rehabilitation in Enschede. RRD performs scientific research on healthcare technology, with a focus on the end user and innovation. Their expertise lies in the areas of rehabilitation, sports, and elderly. The main goal is to develop new technologies or devices that can be used in the clinic or in daily life, to improve the quality of living of their target group or stimulate reintegration into society. One of the research areas of RRD is movement analysis in rehabilitation. In order to gain insight into movements and how these are established, RRD has access to facilities to register movements, external forces and muscle activity.

Currently, research projects are ongoing using clinical 3D gait assessments to gain insight into the walking dynamics of patients with impaired mobility. In the straight line walking test, a total of 42 reflective markers are placed on the patient's body, and infrared cameras register the relative 3D location of these markers. The patient is asked to walk in a straight line, and the positional data are stored. Existing models provide extensive analysis and visualisation of this three-dimensional data. The development of metrics such as pelvic tilt or knee angle throughout the different stages of gait, give great insight into the patient's movement pattern and the possible treatment they would need.

In comparison to daily life tasks, the straight line is a relatively simple task, which does not capture turns for example. Therefore, this task can be extended to the so-called 'Figure of 8 walk test' (F8WT), as can be seen in Figure 1. In this test, patients are asked to walk around two pilons, 1.5 meters apart, each in a different direction. In the motion of turning in both directions, the patient's ability to stay in balance and symmetric becomes clear.

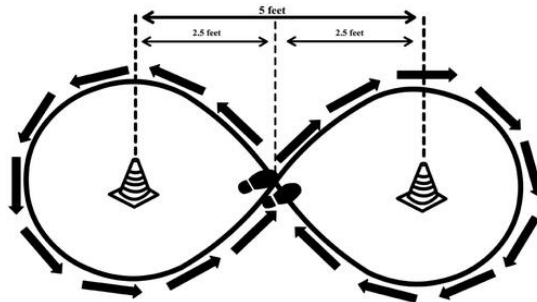


Figure 1: Set up of the Figure-of-8 Walking Test

In the standard F8WT, only the total elapsed time and the total number of steps is tracked, to give an indication of the patient's walking ability. However, extensive analysis of the F8WT may provide new insights in the movement patterns of the subject. Therefore, RRD has conducted experiments of the F8WT in their optical 3D gait analysis environment. Thirty healthy subjects of different age categories are asked to perform the F8WT. The trajectory of their lower body limbs is tracked by placing reflective markers on the body, of which relative locations are stored with a frequency of 100 measurements per second. The result is a very elaborate data set with a wide range of parameters. In this research, a broad and explorative analysis of this data set will be conducted. The aim is to find new aspects of the F8WT that may provide useful information on the walking pattern of a subject, beyond the currently used metrics of total duration and step count.

In this report, first the research questions are formulated in Section 2 and the importance of the research is explained. In Section 3, the available data set and its extension are discussed, and all relevant applied methods are listed and explained. After this, the results and their interpretation follow in Section 4. The report finishes with a discussion and conclusion (Section 5 and 6).

2 Problem

The obtained data set contains three-dimensional positional data of 30 subjects, each performing 3 F8WTs. The goal is to find patterns in the kinematic and spatio-temporal data of the subjects. The main objective is to find clusters of subjects that show similar behaviour on different aspects of gait. This could give a better insight into what gait aspects affect balance and symmetry the most, and could clarify walking dynamics of a curved trajectory in general. However, it is not certain that the desired patterns exist, so this research focuses on exploration of the data set.

2.1 Research question

In line with the research objective, the following research question and sub-questions are posed.

“Can we find patterns in gait deviation from the Figure-of-8 walking test, and how do these patterns influence dynamic balance and symmetry?”

- What gait features, that do not show in the straight line walking test, are introduced in the F8WT and how can they be quantified?
- Do patterns show between these different gait features?
- How can these gait features be clustered, to obtain significantly distinct patient groups?
- How can the related clusters be visualized, and what is their bio-mechanic interpretation?

2.2 Importance of the research

Compared to straight walking tests, the F8WT is a more accurate representation of daily life activities and the complex motor skills they require. It is a valuable combination of straight and curved path walking, and turning gait in both left and right direction. The 3D measurements give a very detailed log of the different gait phases, kinematics, and kinetics of the human body. This can give new insights into gait abnormalities that would not show in the straight path walking test. With an accurate detection of the gait abnormalities, possibly a more accurate rehabilitation can be set up. This could help patients in their rehabilitation process.

2.3 Related work

Gait kinematics in straight path walking have been extensively analysed in many researches, as well as its comparison to curved path walking. Courtine and Schieppati compared body trajectories and gait features of healthy patients in straight and curved line walk [4], [5]. They obtained data using an automatic TV-image processor. The results show that in curved segments, patients shorten the stride length of their inner leg, while the stride length of the outer leg remains the same. The outer leg has higher swing velocity and shorter stance time than the inner leg. The trunk shifts slightly inward, to create a centripetal force to enable the turn, and the feet turn inwards as well.

The comparison between the straight line walking test and F8WT has also been analysed before, even for stroke patients specifically. Using inertial sensors, several spatiotemporal, stability, smoothness, and symmetry-related gait quality metrics can be determined. They show that both healthy patients and patients suffering mildly-severe stroke effects were challenged by the F8WT, while the straight line walking test did not show any impairments [2]. Patients with stroke showed a lower walking speed, increased stride duration, and effected stability in the pelvis [14]. From this, it can be concluded that the F8WT gives additional insights in gait abnormalities, that the straight line walking test does not show.

Most 3D clinical gait analyses with marker tracking only track a straight path, a single corner, or a single pivot turn [6], [7]. The obtained data is used to visualise the trajectory, but no researches have executed 3D measurements on the F8WT with the goal of finding patterns between the gait variables.

3 Methods

Before the data can be analysed, it must first be processed and additional information needs to be computed and added. Then, the specific methods for each stage of the analysis will be explained. All computations and analyses are performed using Python version 3.9.

3.1 Data processing

The data set consists of F8WT trials done by 30 healthy subjects from different age categories (young: age 18-39, middle-aged: age 40-59, and old: age 60-75), with 10 subjects of each category. A total of 42 markers are placed on the patient's torso, pelvis, thighs, shanks, and feet. Each subject is asked to perform the F8WT three times. During the test, the relative locations of the markers are stored with a frequency of 100 measurements per second.

After measurement, a biomechanical model is applied to the data, which returns a wide range of kinematic and kinetic parameters. Examples of these parameters are angles, momentum, and power of the joints. A full overview can be found in the VICON Nexus plug-in gait reference guide [12].

3.1.1 Extension of data set

Before the data can be analysed thoroughly, some important gait kinematics need to be added to the data set. These include the center of mass (CoM) and torso (TOR). The position of these points is expressed as a three-dimensional vector with an x -, y -, and z -direction.

The center of mass is the unique point in space that represents the average location of all mass of the body. This is an intangible point that cannot be measured accurately with the instruments used in the 3D gait analysis. However, research shows that it can be somewhat accurately reconstructed using the position of the sacrum and the two PSIS markers [11]. Therefore, the center of mass is calculated as the average of the 3D-markers of the left and right, anterior and posterior superior iliac spine ('LASI', 'RASI', 'LPSI', 'RPSI'). These vectors can also be seen in Figure 2a, and the center of mass is defined as

$$\begin{pmatrix} x_{CoM} \\ y_{CoM} \\ z_{CoM} \end{pmatrix} = \frac{1}{4} \cdot \left(\begin{pmatrix} x_{LASI} \\ y_{LASI} \\ z_{LASI} \end{pmatrix} + \begin{pmatrix} x_{RASI} \\ y_{RASI} \\ z_{RASI} \end{pmatrix} + \begin{pmatrix} x_{LPSI} \\ y_{LPSI} \\ z_{LPSI} \end{pmatrix} + \begin{pmatrix} x_{RPSI} \\ y_{RPSI} \\ z_{RPSI} \end{pmatrix} \right). \quad (1)$$

In clinical gait analysis, different approximations are used for the torso and its orientation. In this research, we use Plug-in Gait full body modelling with a slight modification [12]. In the original model, the location of the torso is represented by four points in the upper body: C7, T10, sternum and the clavicle. However, in the conducted measurements, the sternum has not been measured. Therefore, we define the torso as the average of the C7, T10, and clavicle only (Figure 2b), to obtain

$$\begin{pmatrix} x_{TOR} \\ y_{TOR} \\ z_{TOR} \end{pmatrix} = \frac{1}{3} \cdot \left(\begin{pmatrix} x_{C7} \\ y_{C7} \\ z_{C7} \end{pmatrix} + \begin{pmatrix} x_{T10} \\ y_{T10} \\ z_{T10} \end{pmatrix} + \begin{pmatrix} x_{CLAV} \\ y_{CLAV} \\ z_{CLAV} \end{pmatrix} \right). \quad (2)$$

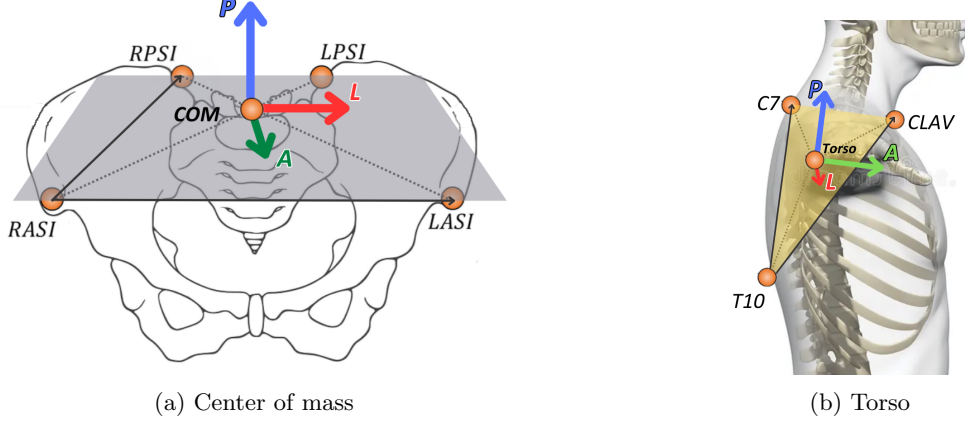


Figure 2: Location and 3D-orientation of Center of mass and Torso

For these two newly defined points, we also compute their orientation. The orientation consists of three orthogonal unit vectors, representing the direction of the anterior (A), proximal (P), and lateral (L) axis of the marker (Figure 3). Each axis has an x -, y -, and z -direction, so $A, P, L \in \mathbb{R}^3$.

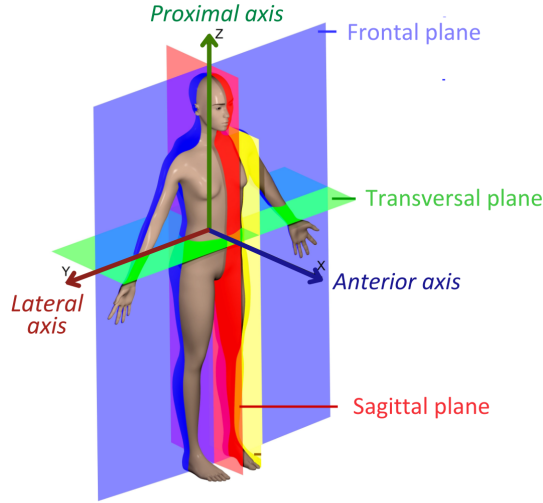


Figure 3: Anatomical planes and axes

The orientation vectors of the center of mass are depicted in Figure 2a and are defined as follows. The direction of the lateral axis, L_{CoM} , is equal to the three dimensional vector from the RASI to LASI marker. The proximal axis, P_{CoM} , is orthogonal to the plane spanned by the lateral axis and the (RASI-RPSI)-vector. The anterior axis, A_{CoM} is orthogonal to the lateral and proximal axes. Orthogonal vectors are computed by use of the cross product, so we have

$$L_{CoM} = \begin{pmatrix} x_{LASI} \\ y_{LASI} \\ z_{LASI} \end{pmatrix} - \begin{pmatrix} x_{RASI} \\ y_{RASI} \\ z_{RASI} \end{pmatrix} \quad (3)$$

$$P_{CoM} = L_{CoM} \times \left(\begin{pmatrix} x_{RPSI} \\ y_{RPSI} \\ z_{RPSI} \end{pmatrix} - \begin{pmatrix} x_{RASI} \\ y_{RASI} \\ z_{RASI} \end{pmatrix} \right) \quad (4)$$

$$A_{CoM} = L_{CoM} \times P_{CoM}. \quad (5)$$

Similarly for the orientation of the torso (Figure 2b), the direction of the proximal axis, P_{TOR} , is equal to the three dimensional vector from marker T10 to marker C7. The lateral axis L_{TOR} is orthogonal to the plane spanned by the proximal axis and the (T10-CLAV)-vector. The anterior axis A_{TOR} is orthogonal to the lateral and proximal axis. This gives

$$P_{\text{TOR}} = \begin{pmatrix} x_{C7} \\ y_{C7} \\ z_{C7} \end{pmatrix} - \begin{pmatrix} x_{T10} \\ y_{T10} \\ z_{T10} \end{pmatrix} \quad (6)$$

$$L_{\text{TOR}} = P_{\text{TOR}} \times \left(\begin{pmatrix} x_{\text{CLAV}} \\ y_{\text{CLAV}} \\ z_{\text{CLAV}} \end{pmatrix} - \begin{pmatrix} x_{T10} \\ y_{T10} \\ z_{T10} \end{pmatrix} \right) \quad (7)$$

$$A_{\text{TOR}} = L_{\text{TOR}} \times P_{\text{TOR}}. \quad (8)$$

3.1.2 Gait and event detection

In each F8WT, there is some redundant data at the beginning and the end of the measurement, in which the subject is not moving. Therefore, data is cut off if the difference in the distance covered by the CoM in the x - y -plane does not lie above a set threshold. This threshold is set at a Euclidean distance difference of 8mm over 0.25 seconds.

If a F8WT trial starts with a left turn around the first pilon, the data is mirrored in the axis through the two pilons, such that all trials start with a right turn.

To be able to compute spatio-temporal parameters, such as the swing time of the leg, the time points of initial contact (IC, or heel strike) and foot-off (FO, or toe-off) need to be detected. Figure 4 shows an overview of these time points and the different phases of a gait cycle.

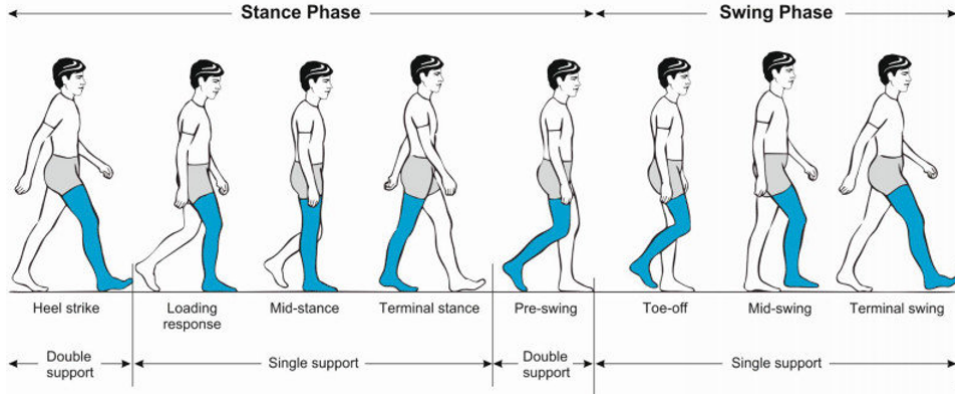


Figure 4: Phases of the gait cycle [9]

The time stamps of IC and FO are detected using the kinematics of the ankle joint center in the z -direction. A moving average function is applied to the data of the ankle joint center to smoothen the trajectories and filter out any noise. This entails that every data point at time stamp t is replaced by the average of the five time stamps from $t - 2$ to $t + 2$. The velocity of the ankle joint center is computed by taking the derivative of the positional data, and similarly the acceleration is derived from the velocity. The trajectory of the ankle joint center is depicted in Figure 5, as well as the velocity and acceleration.

Foot-off (or toe-off) is the point in time where the foot lifts off the floor after the stance phase. These are the points in time where we see a relatively low value in z -position, a positive velocity, and a positive peak in acceleration. Initial contact (or heel strike) is the time stamp where the foot hits the floor again after the swing phase. This is represented by a low z -position, a velocity value close to 0 mm/ds, and a negative peak in acceleration. The first and last step of the subject are not included, as they are not representative for kinematics in regular walking.

An extensive algorithm can be found in Appendix A.1. The results can be seen in Figure 5, where the IC and FO stamps are marked for one F8WT.

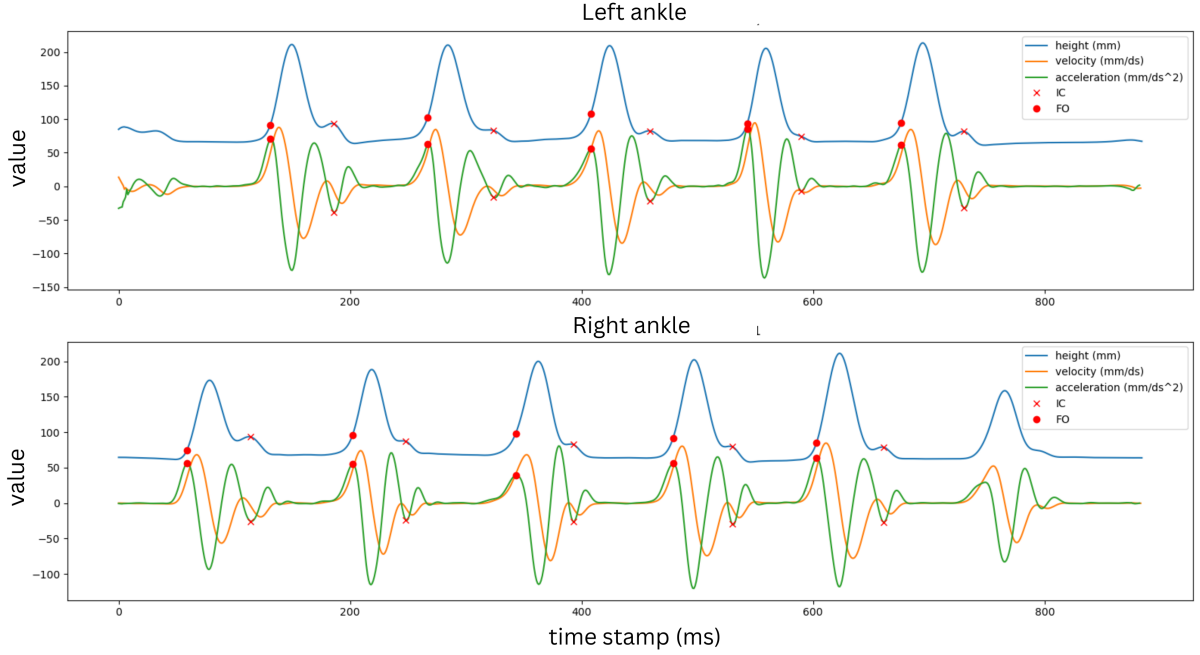


Figure 5: Detection of initial contact and foot-off using the position, velocity, and acceleration of the z -coordinate of the ankle, for the left and right side

3.1.3 Gait feature computation

Based on the positional data, and the IC- and FO-time stamps, a wide range of parameters can be derived. An extensive list of all computed features and their details can be found in Appendix A.2.

The main parameters that are included in this research, are spatio-temporal parameters such as stride/swing/step time and length, the number of steps a subject took, or the total distance they walked. Also, parameters regarding the foot placement are studied, which include the distance and angle between the foot and CoM at initial contact.

Balance and stability are represented by parameters that keep track of the CoM, like the walking velocity throughout the test, and the angular velocity. Lastly, a variety of kinematic parameters are stored, which show the angles of the different joints in the lower limbs.

For each completed F8WT, these parameters are computed and their minimum, maximum, range, mean, and standard deviation are stored. If a gait parameter exists for both the left and right side of the body, a symmetry index is added for each of the five aforementioned statistics. The symmetry index s_p of a statistic p is defined as

$$s_p = \frac{p_L - p_R}{p_L + p_R}, \quad (9)$$

where p_L and p_R are the value for the statistic in the left and right side, respectively. Therefore, a positive symmetry index indicates that the statistic has higher value for the left side, and a negative symmetry index indicates that the right statistic is more dominant. In total, we have a list of 225 gait features, stored for each F8WT individually. With a total of 90 conducted tests, we end up with a feature property data set of 90 rows and 225 columns: $P_{i,f}$, where $i \in [1, 90]$ is a F8WT, and $f \in F$ indicates a gait feature, with $|F| = 225$.

3.2 Clustering of gait features

The total number of 225 gait features is relatively large, making structured analysis challenging. To solve this, the dimension of the dataset should be reduced. The goal is to group together features that are related to each other (Section 3.2.1), resulting in a small number of clusters (Section 3.2.2 and 3.2.4).

3.2.1 Correlation

The correlation coefficient expresses the relation between two variables. In this case, we use the Pearson product-moment correlation coefficient, r , which is a normalized covariance. More specifically, it indicates the strength of a linear dependency between two data sets x and y . We have

$$r_{x,y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (10)$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is the sample mean of x , and similarly $\bar{y}$ is the sample mean of y . The result is a value between -1 and 1 , where -1 indicates a strong negative linear dependency, 1 indicates a positive linear dependency, and values close to 0 indicate a weak or no linear dependency.

For all possible pairs of the 225 gait features, the correlation coefficient C is computed,

$$C_{f_k, f_l} = C_{f_l, f_k} = r_{f_k, f_l} \quad \forall f_l, f_k \in F \quad (11)$$

where f_k and f_l are two gait features. The result is correlation coefficient matrix $C^{225 \times 225}$, symmetric in the main diagonal.

3.2.2 Hierarchical clustering

Given the correlation coefficient of all combinations of gait features, they can be clustered into separate clusters. Each row C_k , $k \in [1, 225]$, in the correlation coefficient matrix C , indicates the correlation between feature f_k and all other features $f_l \in F$. The goal is to create clusters of features that show similar behaviour in the correlation coefficient matrix. To achieve this, a hierarchical cluster analysis (HCA) is applied with Ward's linkage method and Euclidean distance metrics [8]. This method is suitable for data sets with high number of features [1].

For all feature entries (rows) $C_k \in C$, the Euclidean distance d_{C_k, C_l} is measured to all other entries $C_l \in C$,

$$d_{C_k, C_l} = \sum_{i=1}^{225} \sqrt{(C_{k,i} - C_{l,i})^2}. \quad (12)$$

Feature rows with a low total distance d between their correlation values are put in a cluster together, and closely distanced clusters are combined. This process continues until the desired number of clusters, n_{clusters} , is met. For each cluster c_i , the features in that cluster are denoted by

$$F_{c_i} = \{f | f \in c_i, \forall f \in F\}. \quad (13)$$

Each feature is uniquely assigned to one cluster, so we have $F = \bigcup_{i=1}^k F_{c_i}$ and $\bigcap_{i=1}^k F_{c_i} = \emptyset$. This clustering process can be seen in Algorithm 1.

Algorithm 1 Algorithm for computing feature clusters

```
for  $f_k, f_l \in F$  do
    Compute  $C_{f_k, f_l}$  using Equation (10) and (11)
end for
Apply HCA to  $C$  and obtain clusters  $c_1, c_2, \dots, c_{n_{\text{clusters}}}$ 
Calculate silhouette score
```

3.2.3 Quality of clustering

The quality of the distinction made by the clustering can be measured using the silhouette coefficient [10]. The silhouette coefficient s_f shows the relation between the mean intra-cluster distance a , and the nearest-cluster distance b of a feature. The intra-cluster distance of a feature is the mean distance between f and all other features $f' \in F_c, f' \neq f$ in the same cluster. The nearest-cluster distance is the mean distance between feature f and the features from the closest other cluster. The silhouette score of a cluster, s_c , is the mean of the silhouette coefficients of the features in that cluster,

$$a_f = \frac{1}{|F_c|} \sum_{f' \in F_c} |f - f'|, \quad f \in F_c \quad b_f = \min_c \left\{ \frac{1}{|F_c|} \sum_{f' \in F_c} |f - f'| \mid f \notin F_c \right\} \quad (14)$$

$$s_f = \frac{b_f - a_f}{\max(a_f, b_f)} \quad s_c = \frac{1}{|F_c|} \sum_{f \in F_c} s_f. \quad (15)$$

The silhouette score has a value between -1 and 1, where 1 indicates perfect distinct clustering, 0 indicates overlap between clusters, and -1 indicates wrongly assigned samples, which are more similar to another cluster.

3.2.4 Determination of number of clusters

The desired number of clusters can be obtained by applying the clustering method to the correlation matrix C for a range of numbers n_{clusters} , and checking which input gives the highest silhouette score.

An alternative method for finding the optimal number of clusters, is via the gap statistic method [13]. Similarly to previous method, this method iterates over a range of possible number of clusters. The feature rows C_k of correlation matrix C are clustered into a number of clusters k , and the overall within-dispersion measure W_k is calculated. This measure indicates the average within-cluster distance over all computed clusters $c_1, c_2, \dots, c_k$. We have

$$W_k = \sum_{r=1}^k \frac{1}{2|c_r|} \left(\sum_{f_i, f_j \in F_{c_r}} d_{f_i, f_j} \right), \quad (16)$$

where k is the selected number of clusters, $|c_r|$ is the number of features f in cluster c_r , and d_{f_i, f_j} is the Euclidean distance between two features f_i and f_j .

Then, a new reference dataset F^* is created, which is a randomly uniformly distributed copy of the original feature property matrix F . For all 225 features $f \in F$, the minimum and maximum are stored, and 90 new samples are taken from a uniform random distribution. This creates the new reference dataset F^* , with 90 row entries and 225 columns.

From this new reference feature dataset F^* , we construct the reference correlation matrix C^* , apply hierarchical clustering, and calculate the within-cluster distance W_k^* . This is repeated for B random reference datasets that are created and clustered, from which the gap can be calculated with

$$\bar{l} = \frac{1}{B} \sum_{b=1}^B \log(W_{k,b}^*) \quad \text{Gap}(k) = \bar{l} - \log(W_k). \quad (17)$$

The gap indicates the difference between the within-cluster distance of the original data, compared to the average within-cluster distance of randomized copies. In general, the gap will decrease for a higher value of the number of clusters k . The optimal number of clusters is said to be the smallest value for which the gap value still changes significantly. We have

$$s_{k+1} = sd_k \cdot \sqrt{1 + 1/B} \quad (18)$$

$$\hat{k} = \min\{k \mid Gap(k) \geq Gap(k+1) - s_{k+1}\}, \quad (19)$$

with sd_k being the standard deviation of $\bar{l}$. An overview of the full algorithm can be found in Algorithm 2.

Algorithm 2 Algorithm for finding the ideal number of clusters, using the gap statistic method

```

for  $k \in [1, k_{max}]$  do
  Compute correlation matrix  $C$  of  $F$ 
  Apply HCA to  $C$  and obtain clusters  $c_1, c_2, \dots, c_k$ 
  Calculate  $W_k$  using Equation (16)

  for  $b \in [1, B]$  do
    Compute reference feature data set  $F^*$  from  $F$ 
    Compute reference correlation matrix  $C^*$  of  $F^*$ 
    Apply HCA to  $C^*$  and obtain reference clusters  $c_1^*, c_2^*, \dots, c_k^*$ 
    Calculate  $W_{k,b}^*$  using Equation (16)
  end for

  Calculate  $Gap(k)$  and  $s_k$  using Equation (17) and (18)
end for
Find  $\hat{k}$  using Equation (19)

```

3.3 Feature selection

Once all 225 gait features $f \in F$ are divided over separate clusters $c_1, c_2, \dots, c_k$, they are filtered such that only the most important features remain in each cluster.

In this research, the goal is to find gait features that distinguish different gait patterns. Therefore, a feature is of high importance if it contributes greatly to the variance within the dataset. To find these highly variance-contributing features, a principal component analysis (PCA) is applied [3].

For each cluster c_r , we want to find the subset of features $F_{c_r}^* \in F_{c_r}$ that cause the highest variance within the original feature property dataset P . First, P is scaled into P' , using a standard scaler, such that we have zero mean and unit variance. Then, we filter on only the features that are present in the clusters found in Section 3.2.2, to obtain

$$P'_{c_r} = \{P'_{i,f} | f \in F_{c_r}, \forall i \in [1, 90]\}. \quad (20)$$

This filtered set P'_{c_r} contains the scaled results of the features in the selected cluster, for all 90 F8WT's. Principal component analysis is applied to the features, and this returns principal components $p_1, p_2, \dots, p_n$ out of which the dataset P'_{c_r} is composed. For each principal component, we have

$$p_i = \sum_{f_k \in F_{c_r}} a_{f_k,i} \cdot f_k \quad (21)$$

where $a_{f_k,i}$ is the loading of feature f_k in principal component p_i , for $i \in [1, n]$. A high absolute value of the loading means that the feature has high contribution in that principal component, and thus has high contribution to the variance of the dataset.

Each principal component p_i has an explained variance $v(p_i)$, which is a measure of how much of the total variance in the data set is explained by that principal component. We limit ourselves to the first n^* principal components that are cumulatively responsible for the first 75% of the variance within P'_{c_r} ,

$$n^* = \min \left\{ m \mid \sum_{i=1}^m v(p_i) \geq 0.75 \right\}. \quad (22)$$

Then, for each feature $f \in F_{c_r}$, the weighted loading l_f is calculated, consisting of the absolute value of the loading in each principal component, multiplied with the explained variance of that component. We get

$$l_f = \sum_{i=1}^{n^*} |a_{f,i}| \cdot v(p_i) \quad \text{for } f \in F_{c_r}, c_r \in \{c_1, c_2, \dots, c_k\}. \quad (23)$$

The 10 features with the highest absolute weighted loading are chosen as so-called 'cluster representatives', $F_{c_r}^* \in F_{c_r}$, as they have the highest contribution and thus the highest importance within the cluster. All other features are not included in the research, as they do not have significant contribution in distinguishing the walking motion of the subjects.

The result is a number of distinctive clusters $c_1, c_2, \dots, c_k$, each containing 10 features $F_{c_r}^*$ that represent that cluster and have relatively high contribution to the variance in the data set. The feature selection process can be found in Algorithm 3.

Algorithm 3 Algorithm for feature selection after clustering

Compute P' by standardizing P

for cluster $c_r \in \{c_1, c_2, \dots, c_k\}$ **do**

 Compute P'_{c_r} from P' , using Equation (20)

 Apply PCA on P'_{c_r}

 Find n^* using Equation (22)

for $f \in F_{c_r}$ **do**

 Compute l_f , the weighted absolute loading, from n^* using Equation (23)

end for

 Select $F_{c_r}^* \in F_{c_r}$, the 10 features $f \in F_{c_r}$ with the highest weighted loadings l_f

end for

3.4 Clustering of F8WTs

The newly computed ‘cluster representatives’ $F_{c_r}^*$ contribute significantly to the variance of the data set, and thus should be good features to distinguish differences in walking patterns. To validate if this is true, we group the F8WTs on these features $F_{c_r}^*$, and check for distinctive walking patterns.

Similarly to the clustering method in Section 3.2.2, a hierarchical cluster analysis (HCA) is applied. We look at the standardized data set P' , which contains the values of all 225 features for all 90 F8WT’s, standardized to have $\mu = 0$ and $\sigma = 1$ for each feature. For every cluster $c_r \in \{c_1, c_2, \dots, c_k\}$, P' is filtered to only contain the features in $F_{c_r}^*$, so we have

$$P_{c_r}'^* = \{P_{i,f}' | f \in F_{c_r}^*, \forall i \in [1, 90]\}. \quad (24)$$

The HCA with Ward’s linkage method and Euclidean distance metrics is applied, resulting in a unique clustering of the F8WTs into a predetermined number of clusters. The silhouette score measures the distinction of the clustering (Section 3.2.3), indicating the ability of feature set $F_{c_r}^*$ to distinguish different walking patterns. An overview of the cluster method can be found in Algorithm 4.

Algorithm 4 Algorithm for clustering F8WTs based on cluster representative features

```

Compute  $P'$  by standardizing  $P$ 
for cluster  $c_r \in \{c_1, c_2, \dots, c_k\}$  do
    Compute  $P_{c_r}'^*$  from  $P'$ , using Equation (24)
    Apply HCA on  $P_{c_r}'^*$  and obtain clusters of F8WT’s
    Calculate silhouette score
end for

```

The original data set consists of 30 individuals, each performing the F8WT three times. In general, the walking pattern of one individual should be similar throughout the three F8WTs which the individual performs. Therefore, a good distinctive feature set F should group together three similar trials of a single individual into same cluster. If the three trials are put into three different groups, it indicates poor distinction of the feature set.

Additionally, it is interesting to see the similarities between the created F8WT clusters, based on the different feature clusters. If multiple feature clusters make comparable F8WT groupings, it would indicate that a distinctive walking pattern can be defined from the combination of features of those clusters. If no similarities show, it means that every feature cluster has a unique distinctive impact on the F8WTs.

4 Results

4.1 Gait feature clusters

Before the clusters with gait features can be compiled, the optimal number of clusters needs to be found. Then, the clusters are compiled and analysed, and the 10 most important features are extracted per cluster.

4.1.1 Number of clusters

To find the optimal number of clusters, the clustering from Algorithm 1 is applied to a wide range of number of clusters, $n_{\text{clusters}} \in [2, 50]$, and the corresponding silhouette score is calculated. The results are shown in Figure 6. The silhouette score is relatively high for $n_{\text{clusters}} = 2$, and quickly drops down to a value of 0.136 for $n_{\text{clusters}} = 5$. After that, the silhouette score seems to linearly increase for higher n_{clusters} . From this result, no clear optimal number of clusters can be concluded. This could indicate that there is no clear distinction between the gait features.

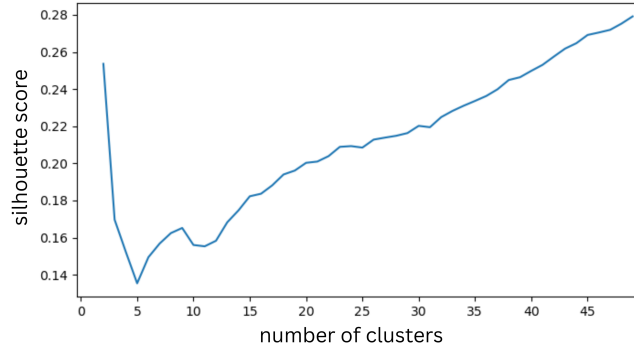
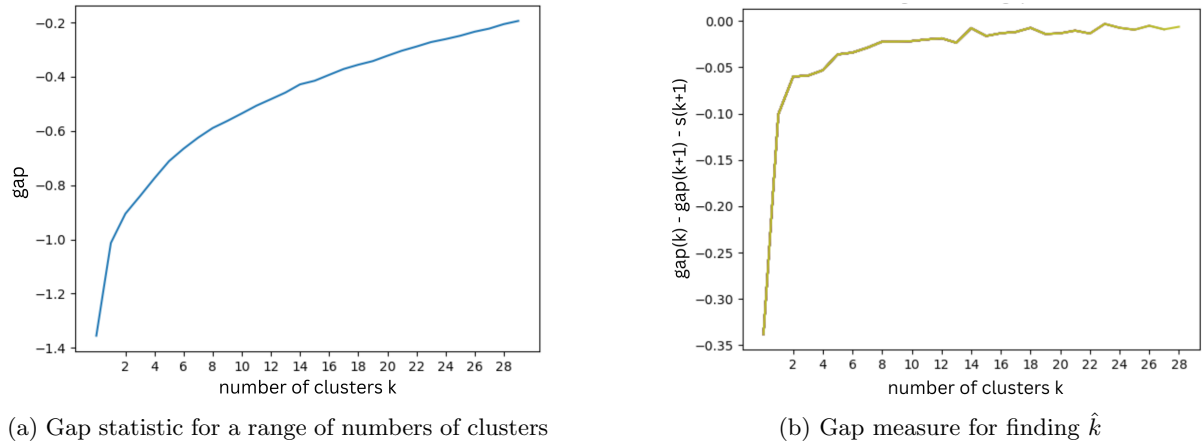


Figure 6: Silhouette score for a range of number of clusters

When applying the gap statistic method, the algorithm also does not seem to converge (Figure 7a), which is in line with the results of the silhouette score test. The gap value still changes significantly for values of $n_{\text{cluster}} = 30$, as can be seen in Figure 7b. The depicted measure is $\hat{k}$ from Equation (19), which is used for determining the ideal number of clusters. A positive value of this measure means that the gap change is not significant anymore, and $\hat{k}$ can be selected. Although the value seems to slowly converge to 0, a value of $n_{\text{cluster}} = 30$ is already too big to be useful in this research, as we are looking for a concise number of clusters. As a result, we choose $n = 6$ as it gives a clear overview of different groups, but is not too large.



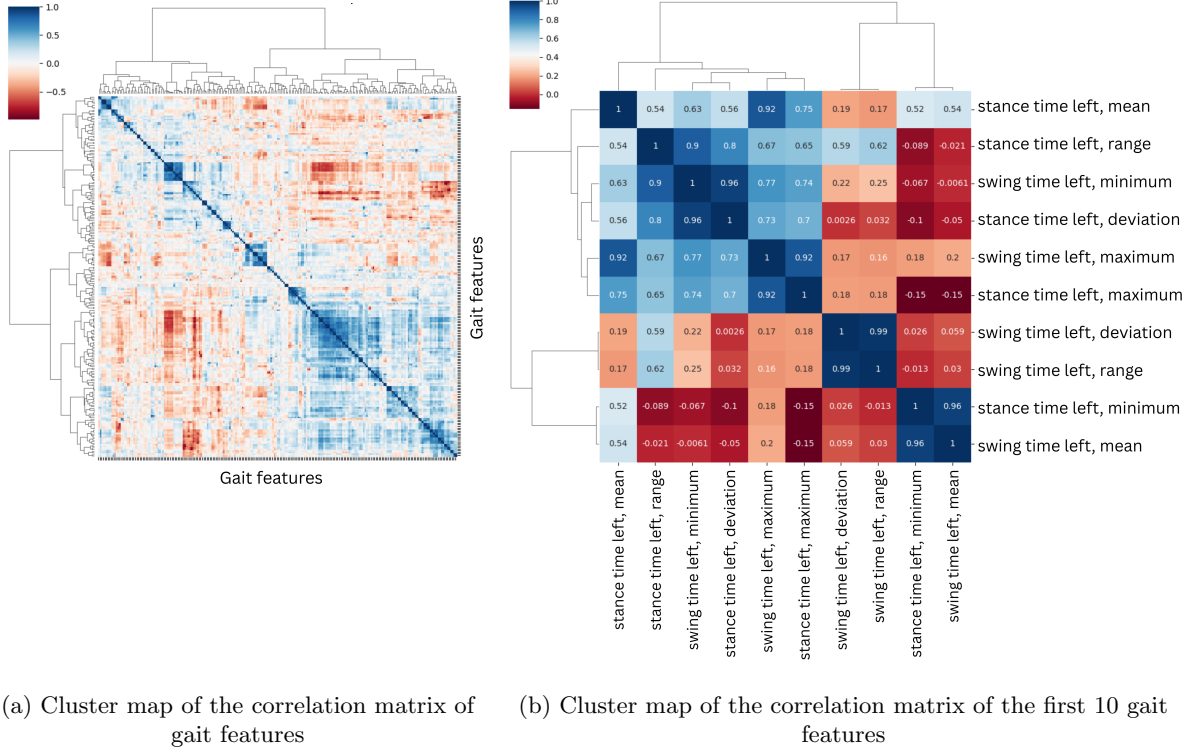
(a) Gap statistic for a range of numbers of clusters

(b) Gap measure for finding $\hat{k}$

Figure 7: Results of the gap statistic method

4.1.2 Cluster analysis

When applying hierarchical cluster analysis, the correlation matrix and cluster division can be visualised in a cluster map, as can be seen in Figure 8a. The cluster map shows the values of the correlation matrix of the gait features, where red is a low value, and blue is a high value. Rows are reordered, such that similar rows, with low Euclidean distance, are positioned underneath each other. The dendrogram on the left shows which rows (gait features) are clustered together, for different stages in the clustering process. Figure 8b shows such a cluster map for the first 10 gait features.



Since $n_{\text{clusters}} = 6$ was set, the algorithm results in a division of all 255 features over 6 clusters, which can be found in Appendix A.3. Figure 9 shows the number of features per clusters, as well as the mean intra-cluster distance (Equation (14)) and mean silhouette coefficient (Equation (15)) of each cluster.

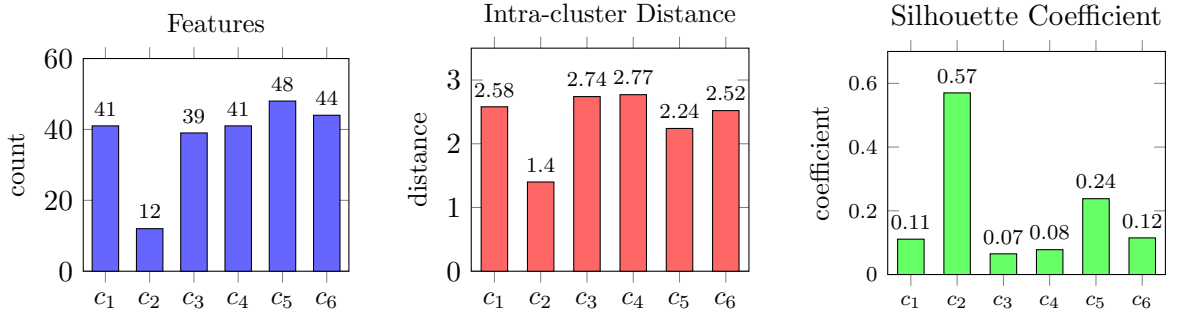


Figure 9: Division of gait features as result of hierarchical cluster analysis

From this result, we see that c_2 has the smallest number of features, the smallest mean intra-cluster distance, and by far the highest mean silhouette coefficient. This indicates that the features in that cluster are highly correlated with each other, and they are very distinct from other features. The clusters with the highest intra-cluster distance and lowest mean silhouette

scores are c_3 and c_4 , which indicates the presence of outliers, or dissimilarities between features in those clusters. Cluster c_5 has the largest number of features, but performs relatively well in its mean silhouette coefficient. This means that the features in this cluster are all relatively similar to each other.

4.2 Feature selection

When applying the principal component analysis, the principal components show different behaviour regarding the explained variance per cluster. This can be seen in Figure 10 and 11.

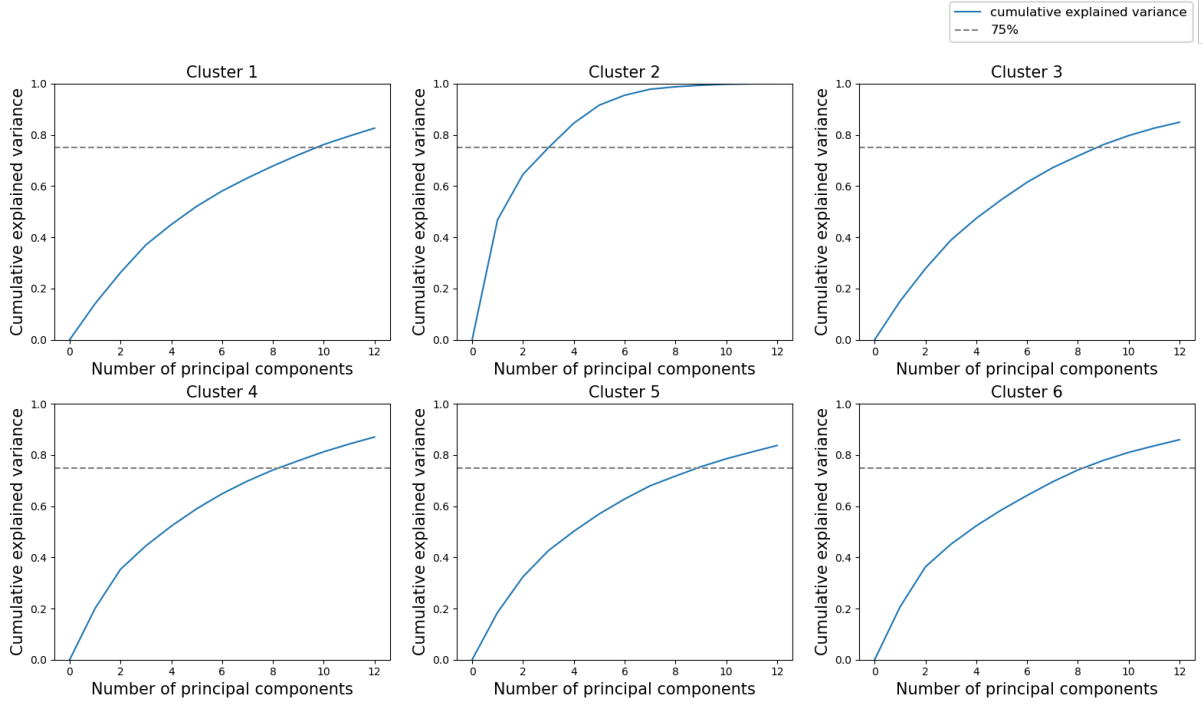


Figure 10: Cumulative explained variance of the principal components for all clusters

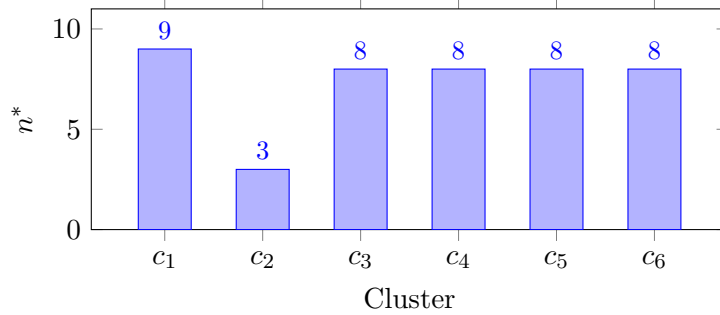


Figure 11: Number of principal components with $\geq 75\%$ of explained variance per cluster

Cluster c_2 has a very high cumulative explained variance in the first principal components (PCs), and already reaches 75% in the first three PCs. This again indicates that the feature entries within the cluster are very similar and highly correlated, and have the same dimensionality.

All remaining clusters have very similar behaviour with respect to each other. The cumulative explained variance is highest for the first PC's, but the slope does not change too much. This means that the features within these clusters have high dimensionality and many differences with each other. It is likely that these clusters are less distinct from each other.

The principal components are used to find the 10 features with the highest weighted loading l_f within their cluster. In Table 1, a small part of the result for cluster c_1 is given. The full list of results can be found in Appendix A.3.

The cluster data is decomposed into their principal components, which are ordered on their explained variance. Within each principal component, the gait features have a loading corresponding to their projection onto that principal component. Gait features with a high positive loading contribute positively to the component, and features with low negative loading contribute negatively to the component. Loadings close to 0 indicate that the feature does not contribute much.

	l_f	PC1	PC2	PC3	PC4
angle left foot - CoM at FO, minimum	1.879	5.280	2.878	-3.120	-2.449
stance time left foot, range	1.864	-4.006	2.614	-2.236	-0.398
distance left foot - CoM orientation, minimum	1.858	-2.826	2.785	-3.347	3.017
stance time left foot, deviation	1.853	-4.068	2.430	-2.048	-0.241
stance time right foot, deviation	1.817	-2.431	0.511	-4.319	6.638
angle left foot - CoM at IC, mean	1.816	6.877	1.284	-2.069	-1.652
frontal hip rotation right, mean	1.792	-5.517	-1.817	1.133	0.601
stance time right foot, range	1.775	-2.414	0.783	-4.262	6.675
frontal hip rotation right, minimum	1.751	-4.371	-2.040	2.218	0.680

Table 1: 10 features with highest weighted loading in cluster c_1 , and their first 4 principal components

When analysing the results, one should look for features with high absolute loading, as they contribute the most to the principal component, and thus distinguish the subject data. Features with the same sign are positively correlated, and features with opposite sign are negatively correlated with each other. In case of cluster c_1 , this would mean that PC1 is mainly represented by the angle between the left foot and the CoM at foot-off, and the frontal right hip rotation. These two features are negatively correlated with each other. In addition, the stance time of the left foot also has a large absolute loading, and is positively correlated with the hip rotation and negatively correlated with the left foot-CoM angle.

The interpretation of this principal component is that it separates F8WTs that have either large outward rotation of the left foot, or large outward frontal rotation of the right hip. The large hip rotation is paired with a longer stance time, while the large foot rotation goes with a shorter stance time. Intuitively, this makes a separation between subjects that make the turn from their feet (with many small steps) or from their hips (with larger steps).

4.3 F8WT clusters

Next, we look at each cluster c_r individually, and validate if the 10 chosen features make distinction in the walking pattern of the subjects. The 90 F8WTs are clustered on their standardized features, into a varying number of F8WT clusters, n_{groups} , and the silhouette score is computed, following Algorithm 4. To not confuse the F8WT clusters with the feature clusters, we call the F8WT clusters ‘F8WT groups’, although they are essentially the same as clusters.

Figure 12 shows that for $n_{\text{groups}} > 5$, cluster c_3 gives the highest silhouette score, indicating that this cluster gives the highest quality grouping. Cluster c_2 performs best for small numbers of groups, and decreases for larger n_{groups} . Clusters c_4 , c_5 , and c_6 perform relatively bad for all numbers of F8WT groups, and seem to converge to the same score for larger n_{groups} . Cluster c_1 performs worst.

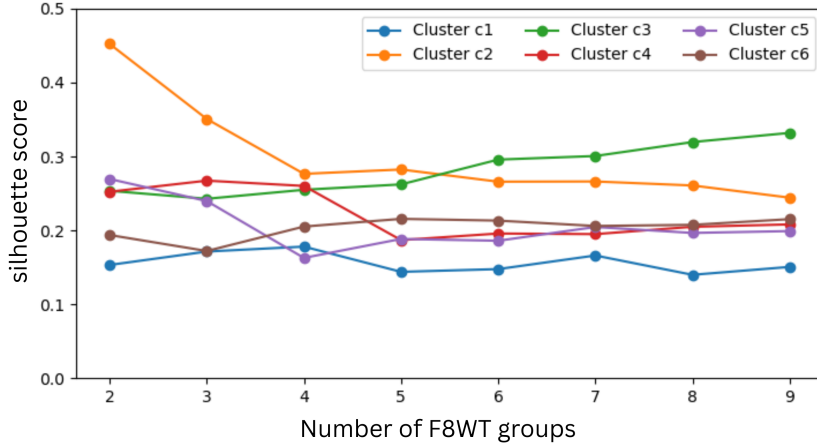


Figure 12: Silhouette score when grouping F8WTs on features from each cluster c_r , for different numbers of F8WT groups

From this result, it seems that cluster c_2 and c_3 give the highest quality clustering, and thus the gait features in those groups make good distinction between the 90 F8WTs. Cluster c_1 performs worst, so the features in this cluster are no good representation of different walking patterns. The remaining three clusters c_4 , c_5 and c_6 perform average.

4.3.1 Division of the three F8WTs per subject

We check for the number of subjects of which all three F8WT trials are put in the same group, and the case where all three trials are put in three distinct groups. This is done for a range of 2 to 10 F8WT groups, $n_{\text{groups}} \in [2, 10]$.

As can be seen from Figure 13a, cluster c_3 puts the three trials in the same group for all but one subject, for all values of n_{groups} . This means that the features in that cluster result in very consistent grouping. In the previous result, cluster c_2 appeared to have high quality clustering, as the silhouette score is relatively high. However, in the F8WT group division, this cluster only manages to put the F8WTs of less than half of the 30 subjects in the same group for $n_{\text{groups}} = 10$. The same relatively bad performance can be seen for cluster c_5 .

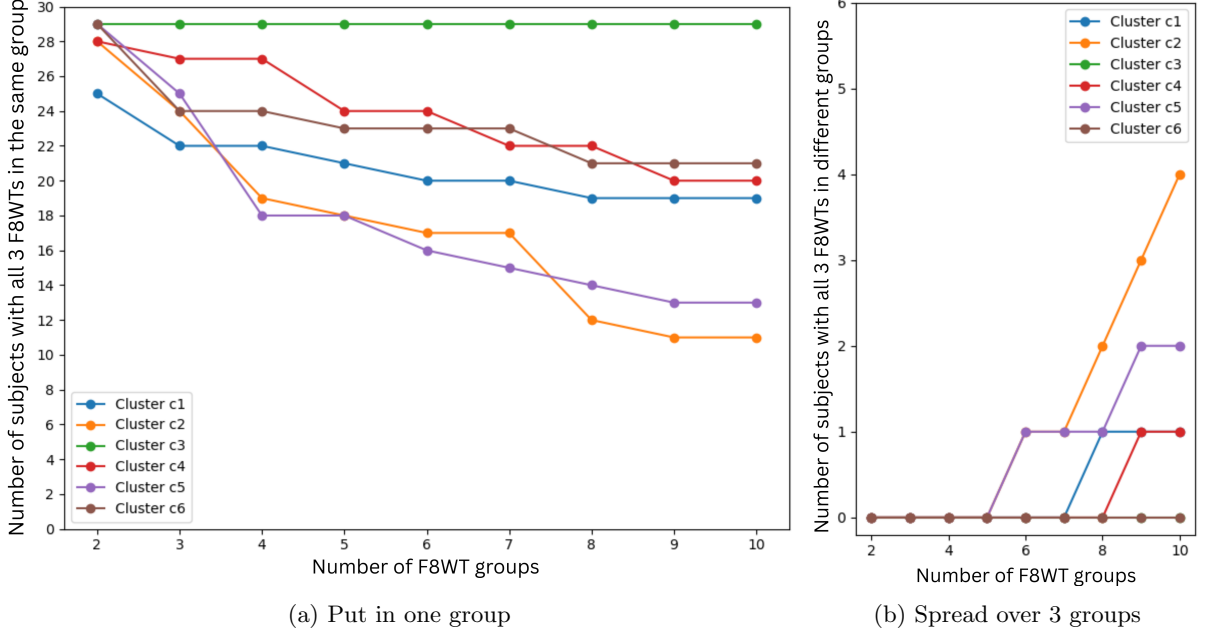


Figure 13: Number of subjects with their three F8WTs put in 1 group, or spread over 3 distinct groups, for different numbers of groups

Figure 13b also confirms the decreasing quality of cluster c_2 and c_5 . If the three F8WTs of one subjects are spread over three distinct groups, it means the selected features do not represent the walking pattern well. For cluster c_2 and c_5 , this occurs even more often as the number of F8WT groups grows.

The created groups can be visualized by the principal components (PCs) of the F8WTs per feature cluster. We apply PCA to the data set filtered on the features only in c_3 , as in Section 3.4. In Figure 14, each data point represents one F8WT, with the value of the first PC on the x -axis, and the value of the second PC on the y -axis. Each subject is given its own color. For clarity, the figure is split up in two, with the first 15 subjects on the left side and the other 15 subjects on the right side. The figure clearly shows that F8WTs with the same color (so of the same subject) group together for cluster c_3 .

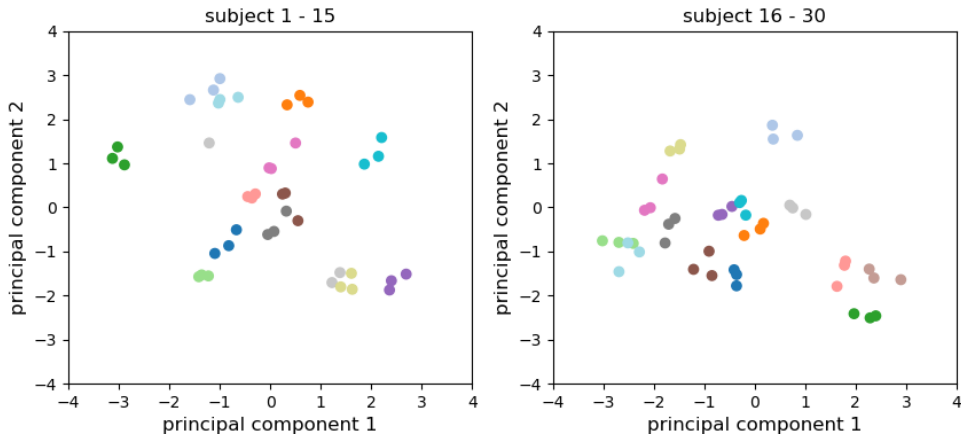


Figure 14: Principal components of the F8WTs with features from c_3 , with a unique color for each subject

We apply the same methods for cluster c_5 , of which the results are shown in Figure 15. This clearly shows that the subjects are more spread throughout the grid, and do not cluster together per subject. Similar results can be seen for the remaining cluster c_1, c_2, c_4 and c_6 , which are shown in Appendix A.4.1.

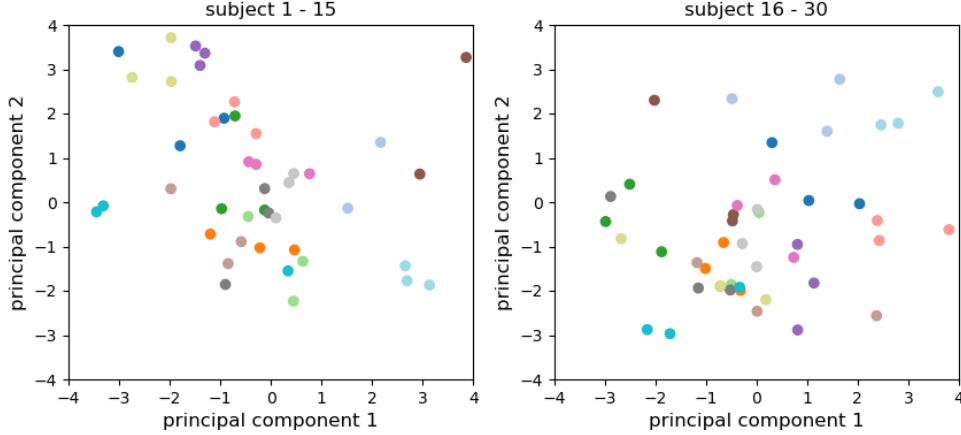


Figure 15: Principal components of the F8WTs with features from c_5 , with a unique color for each subject

Together with previous results, we can conclude that the features in cluster c_3 are very good features to make distinctions in the walking pattern of the subjects. This cluster is mainly represented by features regarding transversal and sagittal rotation of the left ankle, and frontal torso rotation (Appendix A.2). This could mean that left ankle rotation occurs in a variety of different ways, and can be somewhat characteristic of an individual's gait. The features in c_4 , and c_6 are relatively good, and the features in c_1 , c_2 and c_5 perform the worst.

4.3.2 Similarity of groups between clusters

Next, we look for similarities between the created F8WT groups by the different feature clusters.

Since cluster c_3 seems to make an accurate distinction between the different F8WTs, this feature cluster is taken as baseline. The F8WTs are divided over groups for all feature clusters $c_r, r \in [1, 6]$, and the group division is compared with the grouping of c_3 . This can be seen in Figure 16, where the F8WTs are represented by their first two principal components. In cluster c_3 , all F8WTs groups are given their own color. These colors are also used in the plots of the other clusters, to compare their grouping.

Any clear relations are hard to derive from these figures, as the colors are widely distributed throughout the grid. Therefore, the same procedure is applied to the case where the F8WTs are split up in only three groups (colors), which can be seen in Appendix A.4.2. Again, the colors seem to not show any relation with each other.

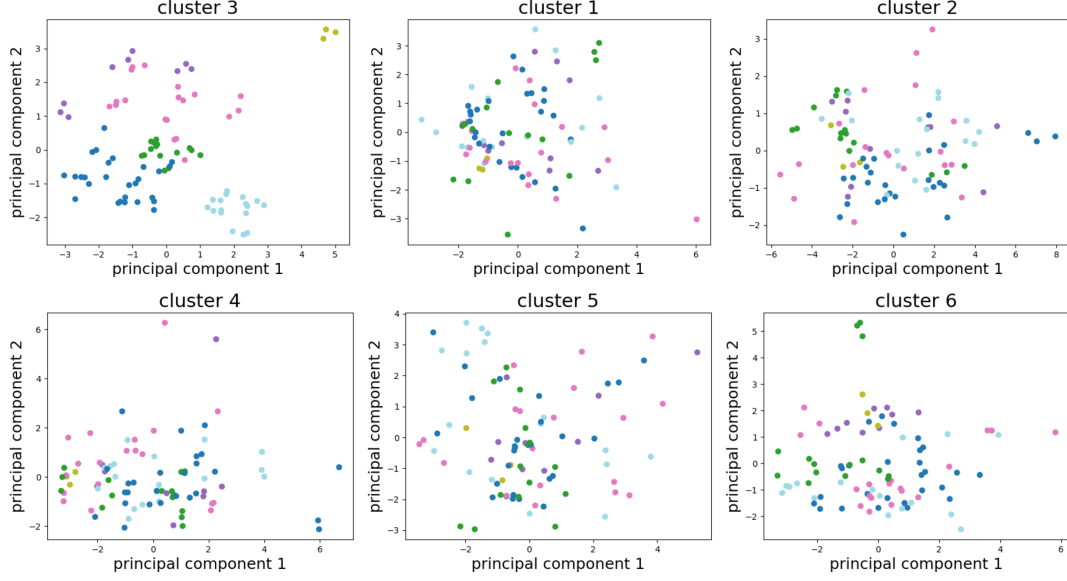


Figure 16: Principal components of F8WTs when grouped into 6 groups, for all 6 feature clusters, with colors according to groups in c_3

To confirm that there are few similarities between groups for the different feature clusters, we compute the silhouette score of the F8WT group labels created by cluster c_i , while filtering the data set on features from another cluster c_j (in Equation (24)). This is done for $n_{\text{groups}} = 6$. The results can be found in Figure 17, and they clearly show that the diagonal entries, with the correct F8WT labels corresponding to the cluster, have the highest silhouette scores. The other entries show scores very close to 0, indicating overlap between the F8WT groups. Some cluster combinations even have very negative silhouette scores, which indicates the existence of incorrectly assigned F8WTs. These F8WTs are more similar to a cluster to which they are not assigned, and therefore should be put in another cluster.

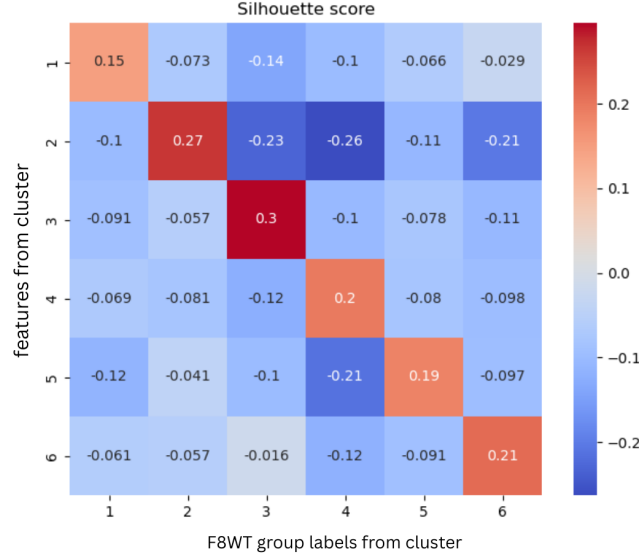


Figure 17: Silhouette score for F8WT groups obtained from cluster c_i , when filtered on features from cluster c_j

From these results, we can conclude that there is very little similarity between F8WT groups computed from the six different feature clusters. This means that every feature cluster has their own unique impact on the grouping of F8WTs.

5 Discussion

The obtained results give some extra insight into the gait features in the F8WT and how they relate to each other. However, there are some limitations to this research.

First of all, only data of healthy subjects is used. In essence, walking behaviour of healthy patients should be relatively similar to each other, so the compiled clusters will not have very clear distinctions. When comparing stroke patient data to the healthy subjects, more clear differences could arise. These differences could give great insight into the details of adapted walking behaviour in stroke patients.

In addition, the data set is not processed extensively before applying the model, which causes some trouble for the applied methods. The reflective markers are labelled using auto-labeling algorithms. These do not give perfect results, so sometimes two closely located markers are accidentally switched. This does not cause too many problems for the location of these markers, as they are just a few millimetres off. However, it can cause problems when these markers are used in the calculation of body angles, as the angle then suddenly can make a 180 degree turn. This greatly influences the minimum/maximum, mean, range, and deviation of the metric throughout the F8WT. Throughout the process, all identified errors were manually corrected before the analysis, but more errors could still be present in the data set.

In future research, more feature statistics should be used in addition to the minimum, maximum, mean, range, and deviation used in this research. These extra statistics should be more robust to data impurities, and should give extra insight in the different stages of the F8WT. For instance, a comparison between the two turning directions and/or the straight line can be analysed, or the differences between the beginning and ending of the turn, for both directions.

Lastly, the model does not give a fully unbiased outcome, as it depends on the chosen input features of the model. The input features are based on three different gait categories from literature: spatio-temporal, balance related parameters, and kinematic parameters. These categories are not mutually independent, and gait features of the same category are more likely to be correlated with each other, and thus to be put in a cluster together. This especially shows in cluster c_2 for example, only consisting of statistics regarding the swing and stance time. Additionally, the three mentioned categories are chosen because they are most common in literature, but other categories could give interesting results as well.

A possible solution is to add more gait features from new categories, although this does not solve the dependency problem completely. In general, gait analysis involves a lot of parameter dependencies. Body parts are connected to each other, and the movement of ligaments is dependent on movement of the other ligaments. Therefore, this problem will always have at least a small contribution.

In addition to the limitations in the inputs of the model, the results could benefit from a more extensive investigation. Only linear correlations have been researched, but more complex correlations may be present in the data set, and could give other insightful results. Additionally, the created clusters and their corresponding results could be investigated more. The current results show to what extent gait features are linearly correlated and how they make different groups in the F8WTs. However, it would also be interesting to gain an insight in how these correlations and groups show in the gait data. For instance, are there any subjects whose gait kinematics clearly show the found correlations, and what does that look like? Or do the different groups of F8WTs show any group-specific properties when performing the test? This would contribute to the clarity of the interpretation of the results, and translate the results to a higher level of detail.

6 Conclusion

In this research, the goal was to find patterns in gait deviation from the Figure-of-8 walking test (F8WT), and how these patterns influence dynamic balance and symmetry.

The answer to this question is found by first introducing balance and stability related measures. These are represented by the displacement of the center of mass, foot placement, and rotation in the transversal plane in general.

The correlations between all relevant gait features are studied, and based on the results, separate feature clusters are created. Although some separation seems to arise between the feature clusters, the overall clustering quality is low. For all but one cluster, there is a low level of distinction within the cluster, and a low level of separation between the clusters. The cluster consisting of only swing and stance time parameters has highest correlation within the cluster and highest distinction with the other feature clusters, but this value is still low.

The feature clusters are used to divide all F8WTs over separate groups. Again, this grouping quality is low. Some feature clusters show consistent grouping behaviour for the three tests performed by one subject. Especially the cluster mainly represented by features regarding rotation of the left ankle, shows consistent and clear distinctive grouping of the F8WTs per subject. This could indicate that left ankle rotation is executed in different ways, and could be characteristic for a subject's walking pattern. Unfortunately, all other feature clusters result in incoherent spread of the F8WTs. Also, no significant similarities can be seen between the F8WT group divisions for different feature clusters.

These results are limited by little variety in the subjects participating in this study, and poor data preparation. Additionally, more insightful results could be obtained by analysing the different segments of the F8WT in more detail, and using more informative descriptive statistics as input parameters for the model. More extensive investigation and visualisation of the obtained features could improve the bio-mechanic interpretation of the outcomes of the model.

To conclude, gait features related to the Figure-of-8 walking test are correlated with each other to such extent, that no significantly distinct clusters can be found. Some separate clusters do arise, but together they fail to differentiate unique walking patterns.

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Appendix

A Appendix

A.1 Algorithm for gait event detection

The algorithm below is used to find the foot-off and initial contact time stamps. Here, the position, velocity, and acceleration of the ankle are denoted by s , v , and a , respectively.

First, the peaks in the z -position value need to be found. These are the points in time with at least a set interval since the last detected peak, z -position above a certain threshold, a peak in the z -position, and at least a set amount of vertical displacement.

Then, the initial contacts are defined as the time stamps where again at least a set time interval (cycle time) has passed since the last initial contact, and also since the last foot-off (swing time). The z -position cannot be larger than a set maximum height. The acceleration has a negative peak, and the velocity is close to 0. Lastly, it should take place within half a swing time after the corresponding detected peak.

The foot-off is the time stamp where again a set time interval (cycle time) has passed since the last detected foot-off time stamp. The acceleration is above a set minimum acceleration value, and it has a positive peak. The velocity is higher than a chosen minimum velocity. Lastly, the foot-off takes place within half a swing time before the corresponding peak.

Algorithm 5 Algorithm to find time stamps of initial contact and foot-off using the z -coordinate of the ankles

$t_{\text{cycle, min}} = 75, \quad t_{\text{stance, min}} = 50, \quad t_{\text{swing, min}} = 20, \quad t_{\text{swing, max}} = 150$

for left and right **do**

$v_{\text{min}} = 0.2 \cdot \max(v)$

$a_{\text{min}} = 0.25 \cdot \max(a)$

$t_{\text{peak interval}} = 15$

$s_{\text{peak, min}} = 0.4(\max(s) + \min(s))$

$s_{\text{max}} = 0.5(\max(s) + \min(s))$

$P = [-\frac{1}{2}t_{\text{cycle, min}}, \text{list of peaks } p_1, p_2, \dots, p_{|P|}]$

 Find peaks in s value:

for $t \in [\frac{1}{2}t_{\text{cycle, min}}, T - t_{\text{cycle, min}}]$ **do**

if $t - p_{|P|} > t_{\text{cycle, min}}$

and $s(t) \geq s_{\text{peak, min}}$

and $s(t-2) < s(t), s(t+2) < s(t)$

and $\sum_{i=t-15}^{t+15} |s(i+1) - s(i)| \geq \frac{1}{2}s_{\text{peak, min}}$ **then**

$P = P \cup t$

end if

end for

$IC = [-t_{\text{cycle, min}}, \text{list of initial contacts } m_1, m_2, \dots, m_{|IC|}]$

$FO = [-t_{\text{cycle, min}}, \text{list of foot-offs } f_1, f_2, \dots, f_{|FO|}]$

for $t \in [0, \min(P_{|P|} + t_{\text{cycle, min}}, T)]$ **do**

 Find initial contact:

if $t - m_{|IC|} > t_{\text{cycle, min}}, t - f_{|FO|} > t_{\text{swing, min}}$

and $s(i) \leq s_{\text{max}}$

and $a(t-2) > a(t), a(t+2) > a(t)$

and $|v(t)| \leq v_{\text{min}}$

and $t > p_{|IC|}, t < p_{|IC|} + \frac{1}{2}t_{\text{swing, max}}$ **then**

$IC = IC \cup t$

end if

 Find foot offs:

if $t - f_{|FO|} > t_{\text{cycle, min}}$

and $a(t) \geq a_{\text{min}}$

and $a(t-2) < a(t), a(t+2) < a(t)$

and $v(t) \geq v_{\text{min}}$

and $t < p_{|FO|}, t > p_{|FO|} - \frac{1}{2}t_{\text{swing, max}}$ **then**

$FO = FO \cup t$

end if

end for

end for

A.2 List of gait features used for analysis

The following gait features are computed and used in the analysis. For each feature, their category is given, as well as the markers and additional data that are used, the applicable side, and the unit of the computed measure. Below the table, an explanation in words is given.

Category	Used data	Gait feature	Side*	Unit
Spatio-temporal	IC	Total number of steps	-	-
Spatio-temporal	IC, FO	Swing time	L, R	cs
Spatio-temporal	IC, FO	Stance time	L, R	cs
Spatio-temporal	IC, Ankle joint center	Stride distance divided by body height	L, R	-
Spatio-temporal	IC, Ankle Joint Centers	Step distance divided by body height	-	-
Spatio-temporal	CoM	Total walked distance	-	mm
Stability	CoM	Walking velocity	-	mm/cs
Stability	CoM, CoM orientation	Body angular velocity	-	degree/cs
Stability	CoM	Distance CoM to pilon	L, R	mm
Foot placement	Foot, CoM, IC	Distance foot to CoM	L, R	-
Foot placement	Ankle-Toe vector, CoM orientation, IC	Foot rotation w.r.t CoM at IC	L, R	degree
Foot placement	Ankle-Toe vector, CoM orientation, FO	Foot rotation w.r.t CoM at FO	L, R	degree
Foot placement	Ankle-Toe vector, CoM orientation, IC, FO	Foot rotation during stance	L, R	degree
Kinematic	Torso angles (S, F, T)**	Torso rotation w.r.t. Pelvis	-	degrees
Kinematic	Hip angles (S, F, T)	Hip angle (pelvis - thigh)	L, R	degrees
Kinematic	Ankle angles (S, F, T)	Knee angle (thigh -shank)	L, R	degrees
Kinematic	Knee angles (S, F, T)	Ankle angle (shank -foot)	L, R	degrees

* L: left, R: right, -: not applicable

** Sagittal plane, Frontal plane, Transversal plane

Table 2: Computed gait features and their properties

Total number of steps: number of initial contacts during the whole F8WT

Swing time: time passed between foot-off and the following initial contact of the same foot (Figure 4)

Stance time: time passed between initial contact and the following foot-off of the same foot (Figure 4)

Normalized stride distance: euclidean distance traveled by the ankle joint center in the x - y -plane between two consecutive initial contacts of the same foot, divided by body height (Figure 18)

Normalized step distance: euclidean distance from the ankle joint center at initial contact to the other ankle joint center at the following initial contact, divided by body height (Figure 18)

Total walked distance: total distance covered throughout the whole F8WT (total time T) by the center of mass in the x - y -plane, with time steps of 5 cs.

$$l_{\text{total trajectory}} = \sum_{t=0}^T \sqrt{(x_{\text{CoM},t+5} - x_{\text{CoM},t})^2 + (y_{\text{CoM},t+5} - y_{\text{CoM},t})^2} \quad (25)$$

Walking velocity: velocity of the three dimensional position of the center of mass

Body angular velocity: angular velocity of the center of mass in the x - y -plane

Distance CoM to pilon: euclidean distance in the x - y -plane between the center of mass and the pilon the body is turning around (Figure 18)

Distance foot to CoM: shortest distance in the $x-y$ -plane between the ankle joint center and the anterior orientation vector of the center of mass at initial contact of the ankle (Figure 18)

Foot rotation w.r.t. CoM at IC: angle in the $x-y$ -plane between the foot and the anterior orientation vector of the center of mass, at initial contact of that foot (Figure 18)

Foot rotation w.r.t. CoM at FO: angle in the $x-y$ -plane between the foot and the anterior orientation vector of the center of mass, at foot-off of that foot (Figure 18)

Foot rotation during stance: difference of angle in the $x-y$ -plane between the foot and the x -orientation vector of the center of mass, between initial contact and foot-off of the same foot.

Torso rotation w.r.t. pelvis: rotation of the torso orientation with respect to the pelvis orientation, in the sagittal, frontal, and transversal plane separately.

Hip angle: rotation of the thigh orientation with respect to the pelvis orientation, in the sagittal, frontal, and transversal plane separately.

Knee angle: rotation of the shank orientation with respect to the thigh orientation, in the sagittal, frontal, and transversal plane separately.

Ankle angle: rotation of the foot orientation with respect to the shank orientation, in the sagittal, frontal, and transversal plane separately.

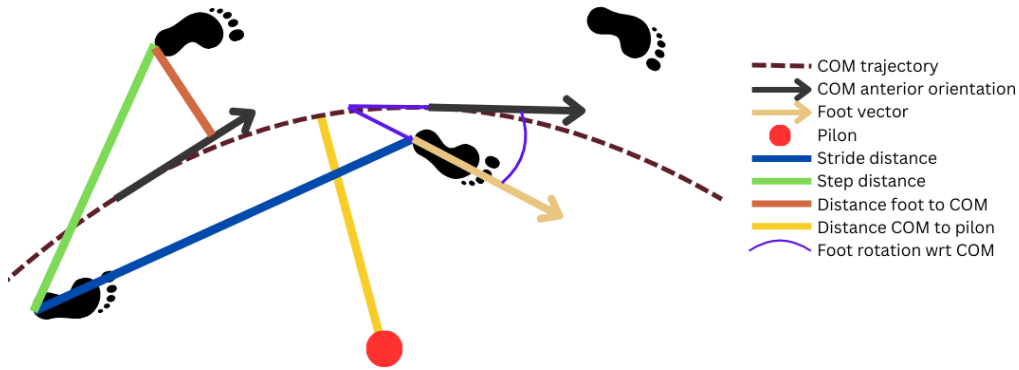


Figure 18: Graphical overview of computed gait features

A.3 Gait feature clusters with principal components and weighted loading

cluster 1	l_f	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8	pc_9
1 angle foot-COM, left, at FO, minimum	1.879	5.280	2.878	-3.120	-2.449	0.692	-0.498	-1.606	1.325	-0.693
2 stance time, left, range	1.864	-4.006	2.614	-2.236	-0.398	-3.064	-2.502	0.620	-2.913	3.980
3 distance foot-COM, left, minimum	1.858	-2.826	2.785	-3.347	3.017	0.998	3.431	-3.163	0.023	1.877
4 stance time, left, deviation	1.853	-4.068	2.430	-2.048	-0.241	-3.289	-2.269	0.974	-3.241	3.984
5 stance time, right, deviation	1.817	-2.431	0.511	-4.319	6.638	-1.091	1.945	2.608	0.306	-1.745
6 angle foot-COM, left, at IC, mean	1.816	6.877	1.284	-2.069	-1.652	-3.083	-1.163	0.091	0.412	-0.545
7 hip rotation, right, frontal, mean	1.792	-5.517	-1.817	1.133	0.601	-1.083	-3.175	1.276	5.735	-0.527
8 stance time, right, range	1.775	-2.414	0.783	-4.262	6.675	-1.210	1.483	1.682	0.104	-1.895
9 hip rotation, right, frontal, minimum	1.751	-4.371	-2.040	2.218	0.680	-0.935	-3.530	1.138	4.700	-0.863
10 ankle rotation, left, frontal, mean	1.728	-2.157	4.916	1.864	-1.495	3.651	-0.285	2.047	-0.829	-2.014
angle foot-COM, left, at IC, maximum	1.710	6.453	0.998	-1.737	-2.129	-2.891	-1.081	0.367	0.199	-0.548
angle foot-COM, left, at IC, minimum	1.702	5.730	1.262	-3.810	-0.190	-1.724	-2.150	-0.785	0.457	-0.017
knee rotation sagittal, symmetry	1.671	2.147	-5.573	-1.625	0.473	2.091	-0.970	-2.713	1.713	1.222
knee rotation, left, frontal, range	1.648	-2.495	-3.100	0.382	-1.784	-3.628	4.826	-0.616	-1.638	-1.887
angle foot-COM, left, at FO, mean	1.640	6.926	1.637	-1.574	0.034	-1.900	-0.512	1.227	1.162	-0.232
knee rotation, left, transversal, minimum	1.600	2.957	1.686	3.784	3.672	1.885	-0.534	-1.119	-0.884	0.291
ankle rotation, right, frontal, mean	1.589	-1.584	6.984	1.589	-2.506	0.330	-0.473	0.859	0.520	-0.616
knee rotation, left, frontal, deviation	1.562	-1.860	-2.846	0.138	-1.379	-4.089	4.964	-1.676	-1.592	-1.890
hip rotation, right, transversal, deviation	1.541	0.823	-2.993	-2.577	-2.350	3.775	-0.291	5.396	-0.133	0.674
ankle rotation, right, frontal, minimum	1.538	-0.647	6.056	0.291	-3.261	0.556	1.943	-1.955	1.682	-2.004
swing time, right, range	1.533	-2.484	-0.561	-2.425	1.397	1.379	-3.978	-2.429	-2.770	-3.506
knee rotation frontal, symmetry	1.528	-2.003	2.052	2.284	-0.774	-3.189	3.314	3.230	1.597	0.535
swing time, right, deviation	1.496	-3.151	-0.276	-2.043	1.275	1.180	-4.007	-1.881	-2.951	-3.189
hip rotation sagittal, symmetry	1.493	1.224	-5.126	-1.377	1.524	0.883	-1.430	-2.997	1.371	1.447
hip rotation, left, frontal, minimum	1.491	-3.065	-2.839	2.346	-2.751	-1.139	-0.679	-1.169	-1.193	0.114
ankle rotation transversal, symmetry	1.483	-1.219	1.964	-2.485	0.719	0.940	3.161	-2.669	4.327	3.497
hip rotation, left, frontal, mean	1.482	-2.545	-3.105	1.041	-3.171	-3.235	-0.812	-1.072	-0.903	0.134
ankle rotation, left, frontal, minimum	1.480	-0.908	5.086	0.623	-2.479	3.377	1.281	-0.501	-0.446	-2.471
knee rotation, left, transversal, mean	1.471	2.606	0.891	4.158	4.271	1.566	0.156	-0.431	-1.072	0.299
knee rotation, left, transversal, maximum	1.380	2.234	0.878	4.474	4.204	1.358	-0.025	0.044	-0.729	-0.107
knee rotation, right, frontal, deviation	1.372	1.773	-5.939	-0.323	-1.180	1.235	1.858	-0.267	-0.865	-0.448
knee rotation, right, frontal, range	1.324	1.408	-5.870	-0.225	-1.193	1.249	1.606	0.588	-0.095	-1.720
angle foot-COM, left, at FO, maximum	1.323	5.041	0.097	-0.363	1.881	-2.988	-0.492	2.246	-0.699	0.545
hip rotation, right, transversal, range	1.272	0.091	-3.466	-1.033	-1.653	3.281	0.614	5.658	-0.805	-0.001
knee rotation, right, transversal, maximum	1.268	1.679	-0.789	6.516	1.337	-0.460	-0.034	0.280	-0.266	1.428
knee rotation, right, transversal, mean	1.247	1.353	-0.840	6.612	1.310	-0.135	0.306	-0.151	-0.915	1.280
walking velocity, minimum	1.148	-2.113	-0.225	2.634	-1.541	-1.044	-1.115	-1.467	2.248	-2.187
knee rotation, right, transversal, minimum	1.112	1.670	0.325	6.110	0.093	0.327	-0.099	-0.557	-1.301	1.145
stance time, left, symmetry	1.101	-3.660	0.036	-0.753	-2.506	0.775	-1.805	-0.104	-0.412	2.606
distance COM-pilon, right, maximum	1.079	0.008	-0.992	-2.727	-1.873	4.130	0.976	-0.827	-1.511	1.116
distance COM-pilon, left, maximum	0.983	-0.758	0.246	-1.716	-0.845	4.521	2.046	-0.177	0.282	2.932

cluster 2	l_f	pc_1	pc_2	pc_3
1 swing time, left, maximum	2.427	3.657	3.719	0.522
2 swing time, right, minimum	2.138	3.709	-1.372	-1.505
3 stance time, left, maximum	1.973	-2.873	-1.339	3.742
4 stance time, right, minimum	1.864	-3.317	0.863	-1.499
5 swing time, right, mean	1.762	2.827	-2.092	-0.634
6 stance time, right, maximum	1.753	-2.973	1.953	-0.128
7 stance time, right, mean	1.724	-3.089	1.224	-0.575
8 stance time, left, minimum	1.711	-2.730	-1.425	-1.725
9 swing time, left, mean	1.688	2.997	1.441	0.266
10 swing time, right, maximum	1.596	2.247	-2.731	0.565
stance time, left, mean	1.466	-2.795	-0.877	-0.007
swing time, left, minimum	1.311	2.341	0.635	0.979

cluster 3	l_f	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8
1 torso rotation, frontal, mean	2.248	-4.406	-2.578	-4.119	-2.472	-1.463	-3.367	2.425	-2.515
2 ankle rotation, left, sagittal, maximum	2.141	2.070	-3.762	4.802	-3.152	1.188	1.793	5.068	1.151
3 knee rotation transversal, symmetry	2.097	-4.312	-3.882	-2.020	-2.949	-2.548	-1.051	2.666	-1.480
4 angle foot-COM, right, at FO, maximum	2.077	5.604	-2.463	-5.002	1.912	-1.049	-0.578	1.300	-0.237
5 ankle rotation, left, transversal, mean	2.020	-3.773	-3.259	1.288	5.392	3.727	-0.736	-0.934	1.234
6 torso rotation, frontal, minimum	1.999	-3.376	-2.576	-3.942	-1.630	-2.281	-3.799	0.782	-2.564
7 ankle rotation, left, sagittal, mean	1.950	2.057	-3.278	6.007	-2.877	0.015	1.409	3.496	-0.353
8 ankle rotation, left, transversal, maximum	1.949	-4.278	-3.943	0.812	4.742	2.977	-0.380	-0.205	1.121
9 ankle rotation, left, transversal, minimum	1.927	-4.344	-2.678	0.741	5.146	3.437	-0.633	-1.892	0.173
10 angle foot-COM, right, at IC, maximum	1.805	5.793	-1.019	-4.848	0.078	2.291	0.061	-0.281	-1.488
ankle rotation, right, transversal, maximum	1.790	-5.112	-0.892	-3.255	0.378	-1.633	5.306	-0.363	-0.335
ankle rotation frontal, symmetry	1.784	4.385	-2.650	4.121	-0.166	-1.067	1.472	-0.518	-2.328
ankle rotation sagittal, symmetry	1.778	1.061	-5.829	4.533	-0.884	-0.178	3.298	1.013	-0.127
torso rotation, transversal, mean	1.776	-4.048	-1.968	-1.460	-3.369	-0.140	-3.779	-0.835	3.360
angle foot-COM, right, at FO, mean	1.724	5.875	-0.449	-3.961	1.540	-1.629	-0.173	0.448	1.147
angle foot-COM, right, at IC, mean	1.714	5.761	-0.205	-4.432	1.041	1.811	-0.054	1.148	0.790
ankle rotation, right, transversal, minimum	1.707	-3.517	1.747	-3.132	0.702	-0.978	4.873	-0.453	-2.644
ankle rotation, right, transversal, mean	1.684	-4.415	0.902	-2.571	0.721	-1.382	5.708	-0.285	-1.192
angle foot-COM, right, at IC, minimum	1.650	5.222	1.169	-2.982	1.476	1.604	-0.050	2.259	0.170
hip rotation transversal, symmetry	1.608	1.392	-1.317	3.829	3.484	1.683	-2.066	-1.741	-3.182
distance COM-pilon, right, mean	1.581	-0.734	5.840	1.564	0.971	2.057	-1.107	3.256	-1.312
total steps, mean	1.574	0.598	4.610	4.319	-1.551	-2.334	-0.115	-1.217	-0.800
foot rotation during stance, right, mean	1.566	1.652	-1.827	1.606	3.770	-5.251	-0.844	-1.918	0.691
angle foot-COM, right, at FO, minimum	1.556	4.846	1.501	-1.959	0.865	-1.656	0.707	-1.104	2.402
foot rotation during stance, left, minimum	1.483	-0.659	5.114	1.082	-1.804	3.802	-2.294	0.374	-0.075
torso rotation, transversal, minimum	1.479	-2.883	-1.236	0.231	-3.308	-1.949	-2.321	-2.504	2.988
angle foot-COM at IC, symmetry	1.429	-1.829	-1.569	2.994	2.960	-2.142	-2.572	0.646	-0.029
ankle rotation, right, sagittal, minimum	1.391	2.657	1.933	1.295	0.440	-1.568	-4.204	-1.447	-1.801
angular velocity, minimum	1.368	0.614	3.245	2.331	-1.927	-0.532	0.608	-4.374	-2.420
distance COM-pilon, right, minimum	1.359	-1.594	5.504	0.707	0.586	0.182	-0.335	2.742	-2.139
distance COM-pilon, left, minimum	1.331	-2.168	5.468	-0.132	2.116	-0.561	0.592	0.206	0.456
distance COM-pilon, left, mean	1.331	-2.442	3.908	0.474	0.227	-0.211	0.054	3.056	4.372
total trajectory length, mean	1.330	-1.802	6.598	0.666	0.373	0.021	0.040	1.438	0.598
foot rotation during stance, right, minimum	1.329	1.578	1.603	1.643	1.220	-4.270	1.004	-0.474	4.139
distance foot-COM, right, mean	1.224	0.594	-0.307	-1.326	-3.935	6.180	-0.266	-1.441	1.241
distance foot-COM, right, minimum	1.141	-0.180	-0.201	-0.717	-4.787	3.779	1.672	-3.481	-0.264
ankle rotation, left, sagittal, minimum	1.131	2.337	-0.003	1.377	-2.285	-1.614	-1.019	-3.288	-1.228
angular velocity, mean	0.873	-0.629	-0.455	-1.069	-2.648	-0.225	1.527	-2.581	2.397
angle foot-COM at FO, symmetry	0.863	2.402	-0.796	0.506	-0.396	1.907	1.621	-0.988	0.085

cluster 4	l_f	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8
1 step distance, right, range	2.300	3.773	4.757	-2.407	-1.346	-2.068	-4.865	0.240	-1.291
2 hip rotation, left, sagittal, mean	2.273	6.432	-4.032	-1.394	-2.011	-0.574	0.265	-0.291	0.398
3 step distance, right, deviation	2.273	3.732	4.821	-2.620	-0.768	-2.392	-4.753	-0.132	-0.984
4 hip rotation, left, sagittal, minimum	2.224	6.520	-4.036	-0.089	-2.209	-0.514	0.829	0.089	-0.815
5 hip rotation, right, sagittal, mean	2.189	5.617	-4.051	0.432	-1.515	1.809	0.378	-1.847	1.215
6 distance foot-COM, left, symmetry	2.186	2.668	4.773	6.467	0.599	-0.366	-0.742	-0.992	3.822
7 swing time, left, range	2.148	0.945	5.289	3.710	-2.922	-1.242	4.719	-1.492	-3.514
8 hip rotation, left, sagittal, maximum	2.144	6.379	-3.331	-1.820	-1.145	-0.173	0.422	-0.977	0.397
9 hip rotation, right, sagittal, minimum	2.132	5.549	-3.841	0.862	-1.790	1.560	0.092	-1.934	0.215
10 hip rotation, left, transversal, deviation	2.126	-4.275	2.814	-3.703	2.404	1.180	3.508	-0.387	-0.221
swing time, left, deviation	2.071	0.874	5.408	3.220	-2.803	-1.210	4.066	-1.689	-3.596
step distance, left, symmetry	2.012	2.449	4.735	-3.662	-3.290	-0.815	-0.891	1.963	-0.094
distance foot-COM, left, mean	2.008	2.041	4.800	4.148	1.138	-0.434	-1.834	-3.683	1.812
stride distance, left, symmetry	2.004	2.437	4.701	-5.221	-0.955	-0.741	-0.722	1.518	1.798
hip rotation, right, sagittal, maximum	1.890	5.635	-2.003	-0.725	-0.412	2.225	-0.201	-2.207	1.959
ankle rotation, right, sagittal, maximum	1.852	0.677	3.307	2.432	3.520	5.861	-1.950	2.021	-2.433
knee rotation, right, frontal, maximum	1.845	-4.780	-1.418	3.218	-1.766	-1.730	-1.140	0.985	-0.126
knee rotation, right, frontal, mean	1.843	-4.207	-2.206	3.786	-1.623	-0.978	-1.081	0.996	0.257
knee rotation, left, sagittal, minimum	1.772	1.851	-4.971	0.059	0.434	-2.931	2.017	4.313	-1.685
hip rotation, right, transversal, mean	1.711	-4.093	-3.168	1.225	0.614	-1.257	-2.004	0.641	0.302
hip rotation, left, transversal, maximum	1.703	-5.281	0.076	-3.719	-0.340	1.599	0.571	-2.243	0.250
hip rotation, left, transversal, mean	1.695	-5.422	-0.088	-3.469	-0.584	1.028	0.244	-2.565	0.457
knee rotation, right, frontal, minimum	1.684	-4.263	-1.848	3.604	-0.805	-0.384	-1.633	0.031	0.714
hip rotation, left, transversal, range	1.664	-3.470	1.165	-3.091	1.820	3.083	2.391	0.126	0.246
hip rotation, right, transversal, minimum	1.654	-3.215	-3.502	1.489	0.483	-1.676	-2.073	1.028	0.359
distance COM-pilon, left, symmetry	1.639	1.039	3.631	-0.109	2.227	-0.085	5.075	3.825	4.714
hip rotation, right, transversal, maximum	1.630	-4.360	-2.769	0.566	0.569	-1.700	-0.846	0.956	-0.591
ankle rotation, right, sagittal, mean	1.596	0.129	2.452	2.904	2.575	5.134	-2.438	1.865	-3.420
knee rotation, left, frontal, maximum	1.587	-5.818	0.224	-0.622	-2.957	0.733	0.065	-0.252	0.832
knee rotation, left, frontal, minimum	1.585	-5.759	-0.478	0.671	-2.914	0.405	-0.238	-0.303	0.357
knee rotation, right, sagittal, maximum	1.554	3.304	-1.557	0.282	3.994	3.968	-0.040	-0.712	0.322
knee rotation, left, sagittal, mean	1.521	1.628	-5.075	-0.637	0.657	-0.454	-0.391	3.220	-2.266
knee rotation, left, frontal, mean	1.518	-5.119	-0.402	0.254	-3.574	0.847	-0.395	0.136	1.047
hip rotation, left, transversal, minimum	1.504	-4.593	-0.065	-2.991	-1.491	-0.032	-0.516	-2.890	0.177
distance foot-COM, right, maximum	1.184	-1.345	0.684	-4.263	2.707	-0.513	-0.998	-0.125	-2.561
step distance, left, deviation	1.144	0.157	-0.780	0.125	6.495	-5.015	0.359	-2.398	0.043
step distance, left, range	1.144	-0.142	-0.650	0.101	6.532	-5.008	0.616	-2.234	-0.397
hip rotation frontal, symmetry	1.130	0.473	3.618	0.373	0.224	-0.393	-0.422	4.590	3.536
foot rotation during stance, left, symmetry	1.106	2.184	1.525	0.243	-0.642	-0.835	2.770	1.349	-1.817
knee rotation, right, sagittal, minimum	1.026	-0.343	-5.076	0.191	-0.332	0.252	1.719	0.331	0.141
knee rotation, right, sagittal, mean	0.960	-0.002	-3.437	0.179	1.205	3.835	0.067	-0.871	0.439

cluster 5	l_f	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8
1 stride distance, right, range	2.633	-5.769	6.692	0.658	2.833	0.868	-0.872	-3.251	-2.004
2 stride distance, right, deviation	2.366	-5.679	5.199	0.206	2.028	1.084	-1.486	-3.608	-1.881
3 stride distance, left, range	2.125	6.613	-2.124	2.637	1.400	-0.582	1.497	-0.859	-1.594
4 ankle rotation, left, transversal, range	2.087	-5.106	0.731	4.102	1.746	-2.917	1.814	2.214	-1.938
5 stride distance, left, deviation	1.881	6.187	-1.933	1.971	0.484	-1.032	1.053	-0.946	-1.311
6 distance foot-COM, left, range	1.848	3.796	5.056	-1.645	-0.154	0.623	1.678	2.297	-0.113
7 distance foot-COM, left, deviation	1.826	3.243	4.551	-2.925	-0.213	0.863	1.491	2.335	-0.343
8 distance foot-COM, right, range	1.822	3.927	4.767	-1.423	-0.593	-0.486	1.446	1.077	-1.841
9 ankle rotation, left, transversal, deviation	1.818	-5.854	0.513	2.801	-0.122	-2.756	1.758	1.116	0.597
10 stride distance, left, minimum	1.805	-4.175	-2.043	-2.511	1.687	3.240	0.034	2.356	-0.572
torso rotation, transversal, maximum	1.797	2.174	2.716	2.066	-1.390	5.368	2.444	-3.144	0.850
step distance, left, minimum	1.755	-2.415	-1.854	-2.652	2.832	4.200	-1.574	3.046	0.847
distance foot-COM, right, deviation	1.750	3.642	4.267	-1.601	-0.929	-0.646	1.313	1.012	-2.014
distance foot-COM, left, maximum	1.701	3.846	4.028	-1.564	0.701	1.024	0.466	-0.004	3.170
stride distance, right, maximum	1.634	-5.715	0.842	-1.541	-0.425	-0.559	1.218	-3.059	-0.101
swing time, left, symmetry	1.604	-0.964	4.925	0.251	4.261	-1.576	0.432	3.972	1.465
ankle rotation, left, sagittal, deviation	1.539	-3.756	0.653	2.074	-1.742	-1.659	-2.418	1.651	1.884
stride distance, right, mean	1.531	-3.971	-1.813	-1.780	-1.803	-0.299	1.699	-1.695	0.509
stride distance, left, maximum	1.528	4.185	-4.007	-0.027	1.113	0.282	0.488	0.093	-1.512
ankle rotation, right, sagittal, deviation	1.462	0.198	-2.188	4.113	-2.523	3.272	-2.006	1.485	-2.530
step distance, right, minimum	1.459	-3.021	-1.895	-0.539	-1.920	1.231	5.055	0.209	1.256
ankle rotation, left, sagittal, range	1.427	-3.358	0.671	1.964	-2.283	-0.039	-2.932	1.949	1.729
stride distance, right, minimum	1.400	-1.739	-4.230	-0.757	-1.597	-0.052	3.901	0.122	1.446
step distance, right, mean	1.370	-2.945	-2.775	-1.727	-1.104	0.721	1.798	-0.414	0.125
torso rotation, frontal, maximum	1.357	1.097	0.107	4.194	2.929	3.307	1.811	-0.466	3.593
ankle rotation, right, sagittal, range	1.339	-0.633	-0.432	3.279	-2.476	4.197	-3.310	1.775	-1.867
hip rotation, right, sagittal, deviation	1.319	2.068	-2.447	-2.015	3.506	-0.023	-1.846	-0.219	0.134
hip rotation, right, sagittal, range	1.305	2.577	-1.689	-1.194	3.207	-1.037	-1.905	0.377	0.748
step distance, left, mean	1.244	-1.357	-3.551	-2.411	0.785	1.164	-1.424	0.474	-0.209
hip rotation, left, frontal, maximum	1.216	2.986	0.223	3.256	-0.170	2.653	-0.338	-1.652	0.081
ankle rotation, right, transversal, range	1.205	0.430	-1.398	5.960	1.086	-2.120	0.396	0.534	-1.245
ankle rotation, right, transversal, deviation	1.195	0.351	-1.826	5.185	0.644	-2.535	1.407	0.204	0.897
hip rotation, left, sagittal, deviation	1.150	-1.106	-2.023	-2.379	3.092	-1.196	0.343	-0.831	-1.154
hip rotation, left, sagittal, range	1.049	0.647	-2.122	-2.274	2.504	-1.847	-0.148	-0.508	-1.390
hip rotation, right, frontal, range	1.041	2.272	-0.853	0.060	1.922	-1.310	-1.612	-1.819	1.960
hip rotation, left, frontal, range	1.019	2.750	0.435	0.582	-0.785	-1.262	-2.204	-1.501	1.049
stride distance, left, mean	0.948	-0.297	-4.055	-1.925	0.268	0.807	-0.244	0.096	-1.017
walking velocity, deviation	0.817	0.873	0.924	-1.877	-1.730	-1.553	-0.661	-1.179	-0.010
angular velocity, maximum	0.816	0.795	1.469	-0.517	-3.015	-1.077	-0.984	0.887	0.208
hip rotation, left, frontal, deviation	0.778	1.691	0.081	-0.060	-1.270	-0.750	-2.149	-1.684	2.338
walking velocity, mean	0.740	-0.891	-0.682	-2.797	-1.409	-0.447	0.363	0.539	-0.237
angular velocity, range	0.731	0.859	0.885	-0.655	-3.141	-1.136	-0.849	0.316	-0.047
walking velocity, range	0.672	0.687	0.164	-1.461	-2.449	-1.515	-0.659	-0.655	-0.355
walking velocity, maximum	0.670	0.633	0.283	-1.298	-2.779	-1.237	-0.779	-0.609	-0.280
angular velocity, deviation	0.653	0.050	0.860	-0.867	-3.529	-1.320	-0.650	0.648	0.213
step distance, right, maximum	0.641	-0.429	-2.427	-1.001	-0.326	0.216	0.046	-0.998	-0.752
hip rotation, right, frontal, deviation	0.639	0.628	-0.322	-0.076	0.884	-1.627	-2.608	-1.424	1.769
step distance, left, maximum	0.611	-0.026	-2.352	-1.860	-0.032	-0.527	-0.289	-0.259	-0.553

cluster 6	l_f	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8
1 ankle rotation, right, frontal, maximum	2.306	-4.800	3.619	4.385	1.883	1.744	0.548	0.834	-0.919
2 angle foot-COM, left, at IC, range	2.292	-3.937	-5.630	-0.110	-0.423	4.937	0.797	-3.206	0.730
3 ankle rotation, left, frontal, maximum	2.270	-3.493	4.186	4.616	1.823	2.987	-1.325	0.984	-0.921
4 angle foot-COM, left, at IC, deviation	2.238	-4.163	-5.276	-0.054	0.483	4.526	0.398	-3.248	-0.756
5 angle foot-COM, right, at IC, range	2.196	-3.769	-4.637	2.204	0.752	1.051	1.151	4.369	1.769
6 torso rotation, sagittal, range	2.179	5.308	-0.761	-3.705	1.901	1.848	-2.006	2.311	-3.244
7 ankle rotation, left, frontal, range	2.147	-3.261	6.603	-1.211	1.159	1.350	-1.404	-1.060	0.654
8 angle foot-COM, right, at IC, deviation	2.081	-3.354	-5.204	2.808	1.384	0.218	0.302	3.423	0.330
9 knee rotation, right, transversal, range	2.062	3.940	3.410	-2.125	-1.475	4.681	0.273	0.536	1.813
10 knee rotation, right, transversal, deviation	2.050	3.239	4.082	-2.874	-2.062	2.674	1.394	0.820	-1.121
ankle rotation, left, frontal, deviation	2.029	-3.017	6.675	-0.792	0.996	0.841	-1.353	-1.308	0.472
ankle rotation, right, frontal, range	1.935	-3.532	5.674	1.160	2.044	-0.012	-0.546	-0.330	0.409
torso rotation, frontal, range	1.920	5.234	-0.444	-2.088	1.677	-1.837	-2.102	1.866	2.890
ankle rotation, right, frontal, deviation	1.910	-3.355	5.537	1.342	1.836	-0.390	-0.809	-0.309	0.292
torso rotation, sagittal, maximum	1.881	4.283	-2.712	2.574	-2.090	0.094	-2.334	0.995	-0.121
torso rotation, sagittal, deviation	1.867	4.629	-0.086	-3.860	1.769	1.221	-1.442	1.671	-3.984
torso rotation, sagittal, mean	1.829	3.702	-2.643	3.662	-2.468	-0.013	-1.800	0.611	0.364
torso rotation, frontal, deviation	1.820	4.013	-1.191	-1.726	1.985	-1.642	-3.068	1.854	2.955
distance COM-pilon, left, deviation	1.813	-2.617	-3.493	-3.468	3.282	-0.437	-1.425	-0.655	-0.908
distance COM-pilon, right, range	1.800	-3.007	-3.527	-3.405	0.988	0.232	3.433	0.427	0.577
knee rotation, left, sagittal, deviation	1.780	4.515	1.479	2.103	-0.149	-1.941	3.233	-1.687	0.609
torso rotation, sagittal, minimum	1.774	2.928	-2.813	4.272	-2.509	0.107	-1.714	0.346	1.094
distance COM-pilon, right, deviation	1.758	-3.769	-2.719	-3.166	1.186	-1.078	1.944	-0.206	0.060
knee rotation, left, sagittal, range	1.714	5.920	-0.636	0.153	0.947	-0.752	3.665	-0.780	-0.363
distance COM-pilon, left, range	1.671	-2.101	-3.733	-3.326	2.917	-0.642	-1.369	-0.360	-0.305
foot rotation during stance, right, maximum	1.644	-4.355	1.418	-0.054	0.381	-0.649	3.045	4.734	0.620
foot rotation during stance, left, range	1.644	-3.620	-1.396	-1.865	-3.333	-0.358	-1.297	-1.354	2.302
hip rotation, right, frontal, maximum	1.611	4.517	0.894	0.440	3.082	-0.826	0.576	-1.532	2.440
knee rotation, right, sagittal, deviation	1.568	3.622	1.907	0.967	-1.173	-0.168	4.499	-0.841	0.959
foot rotation during stance, right, range	1.530	-3.802	-0.811	0.670	-1.623	-2.622	1.905	2.428	-0.918
knee rotation, left, sagittal, maximum	1.521	3.075	0.320	1.263	4.880	-0.899	0.171	-2.275	4.042
knee rotation, right, sagittal, range	1.513	5.544	0.079	0.313	-0.170	-0.059	4.159	-1.105	-0.464
angle foot-COM, right, at FO, deviation	1.480	-2.509	2.517	-0.979	-1.242	-3.040	0.575	-1.034	-2.481
knee rotation, left, transversal, deviation	1.432	2.832	1.889	-0.574	-3.279	1.673	0.487	1.410	-1.257
angle foot-COM, right, at FO, range	1.428	-2.006	2.581	-1.565	-1.713	-3.134	0.954	-0.161	-1.945
torso rotation, transversal, range	1.416	1.844	-1.516	3.921	1.009	-1.753	-1.515	-1.384	-2.412
foot rotation during stance, right, deviation	1.349	-3.691	-1.095	0.224	-0.810	-3.355	0.311	0.818	-1.435
foot rotation during stance, left, deviation	1.339	-2.867	-0.859	-1.423	-3.411	-0.356	-1.510	-1.959	0.673
foot rotation during stance, left, mean	1.334	2.277	-2.153	0.657	1.585	-1.018	-0.062	-2.834	-2.972
angle foot-COM, left, at FO, range	1.334	-1.894	1.594	-1.755	-2.943	-2.084	-1.858	0.284	1.667
torso rotation, transversal, deviation	1.309	1.469	-1.119	3.975	0.809	-1.525	-1.998	-1.531	-2.875
knee rotation, left, transversal, range	1.296	2.978	0.634	0.374	-3.118	3.174	-0.598	1.659	-0.098
foot rotation during stance, left, maximum	1.252	-1.977	-2.127	-0.163	-3.607	-0.671	-0.344	-2.631	0.793
angle foot-COM, left, at FO, deviation	1.136	-0.975	1.483	-1.795	-3.160	-2.096	-1.942	-0.587	0.985

A.4 Clustering of F8WT's

F8WT	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
cluster_1	3	3	3	3	3	3	5	2	2	5	6	5	3	3	3	3	3	3	2	2	2	6	6	6	1	2	1	6	3	3
cluster_2	6	6	6	4	4	6	3	3	3	1	1	1	1	1	1	3	1	3	4	4	4	5	5	5	3	1	1	1	1	1
cluster_3	1	1	1	3	3	3	3	3	3	3	3	3	1	1	1	2	2	2	6	6	6	4	4	4	5	5	5	4	4	4
cluster_4	6	6	6	1	3	3	6	6	6	4	6	6	5	5	5	6	6	6	1	6	6	1	1	4	3	3	3	3	1	1
cluster_5	3	3	3	5	2	5	4	6	4	3	5	5	6	6	6	3	3	3	3	3	3	2	2	2	6	6	5	3	6	6
cluster_6	6	6	6	6	6	6	3	3	3	6	6	6	4	6	6	1	1	1	2	2	2	5	5	5	3	3	3	3	3	3

F8WT	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59
cluster_1	5	1	1	4	2	2	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	5	5	5	2	2	2	6	3	6
cluster_2	1	3	3	1	6	6	5	5	4	2	2	2	4	6	5	6	6	6	6	6	6	1	1	1	6	6	6	1	1	1
cluster_3	2	2	2	4	6	6	6	6	6	4	4	4	4	4	4	1	1	1	4	4	4	1	1	1	6	6	6	1	1	1
cluster_4	6	6	6	6	6	6	1	1	1	3	3	3	3	3	3	1	1	1	2	2	2	1	1	1	3	3	3	6	6	6
cluster_5	6	6	6	6	5	5	3	3	3	6	4	4	1	1	1	5	5	5	3	5	5	6	6	6	4	4	4	6	4	6
cluster_6	2	2	2	4	4	4	5	5	6	4	4	4	4	6	6	3	3	3	4	4	4	4	4	4	4	4	4	2	4	4

F8WT	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
cluster_1	3	6	3	2	2	2	2	2	2	6	6	6	2	2	2	5	5	5	4	2	2	5	5	5	3	2	3	2	2	2
cluster_2	1	6	6	6	6	4	6	5	6	3	3	3	1	1	1	1	1	1	1	3	3	3	3	1	3	2	2	4	4	4
cluster_3	6	6	6	2	2	2	1	1	1	6	6	6	1	1	1	1	1	1	2	2	2	4	4	4	2	2	2	1	1	1
cluster_4	1	1	1	6	6	6	1	3	3	6	6	6	1	1	1	6	6	6	2	2	2	6	6	6	3	3	3	6	6	6
cluster_5	1	1	1	6	6	6	6	3	6	6	4	6	5	6	4	4	4	6	6	6	6	6	4	6	4	4	4	2	2	2
cluster_6	4	4	4	6	6	6	4	4	6	2	2	2	4	4	6	4	4	6	2	2	2	6	6	6	6	6	6	2	2	2

Figure 19: Division of F8WTs into 6 clusters, for each cluster independently

A.4.1 Spread of F8WTs per subject

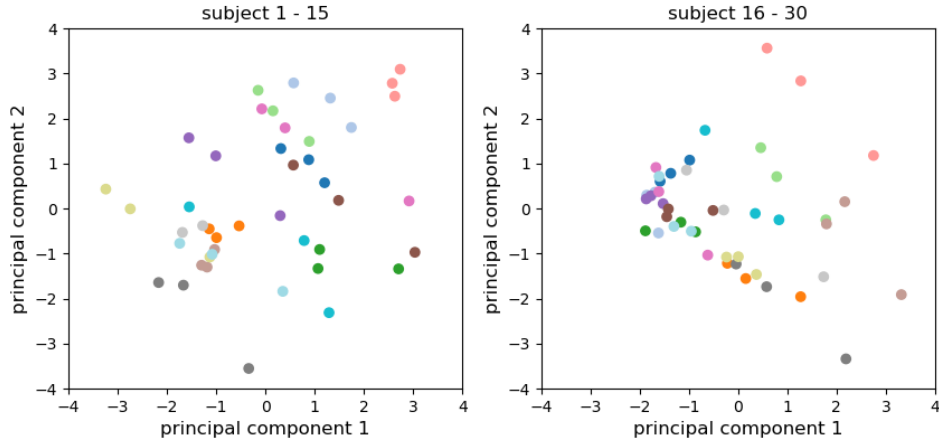


Figure 20: Principal components of the F8WTs with features from c_1 , with a unique color for each subject

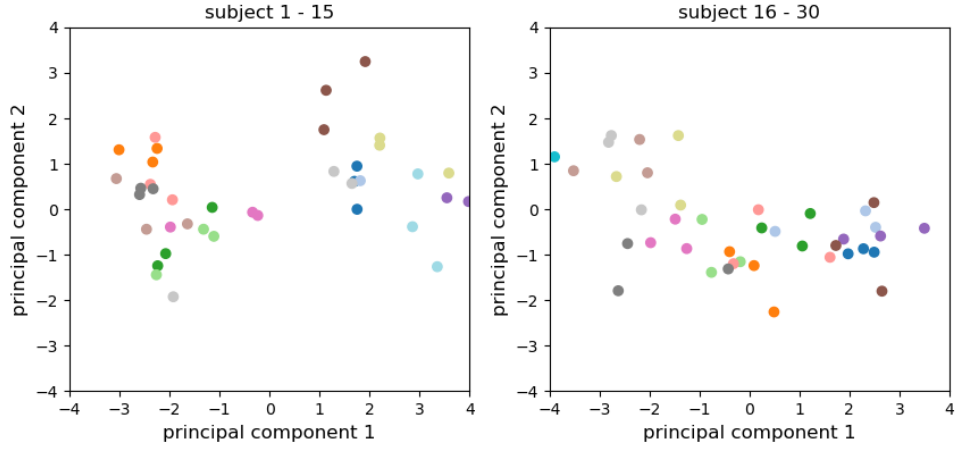


Figure 21: Principal components of the F8WTs with features from c_2 , with a unique color for each subject

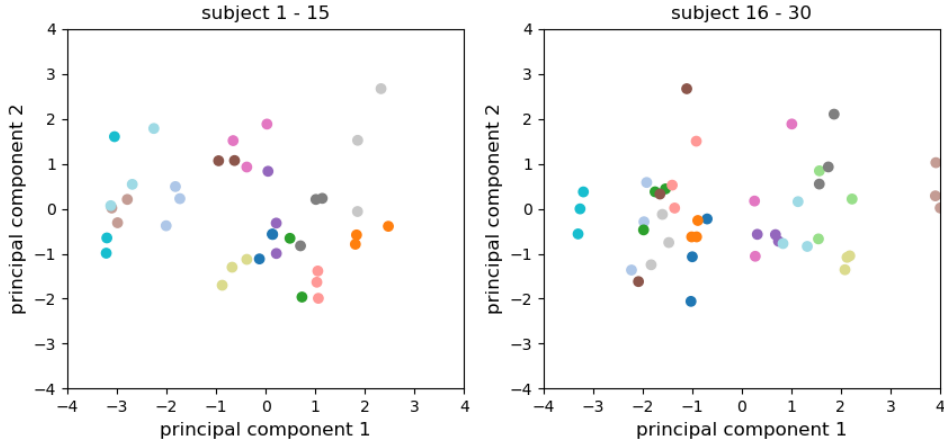


Figure 22: Principal components of the F8WTs with features from c_4 , with a unique color for each subject

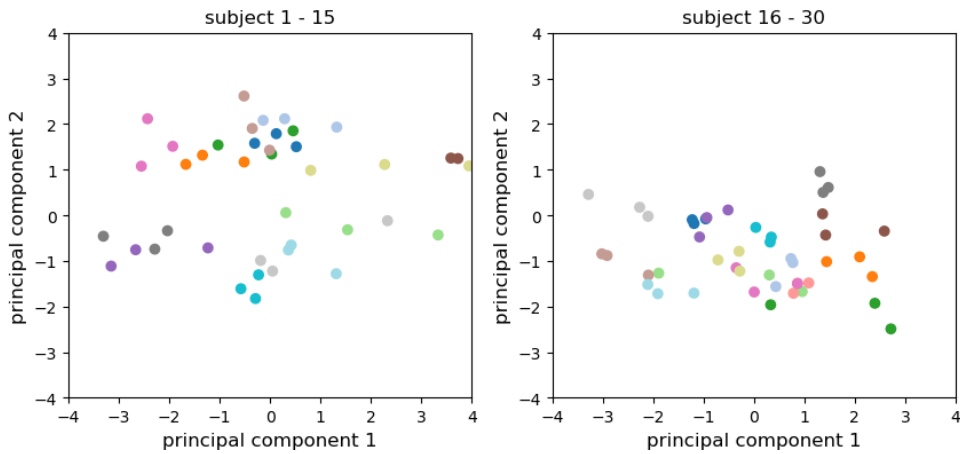


Figure 23: Principal components of the F8WTs with features from c_6 , with a unique color for each subject

A.4.2 Principal components of F8WTs, grouped into 3 groups

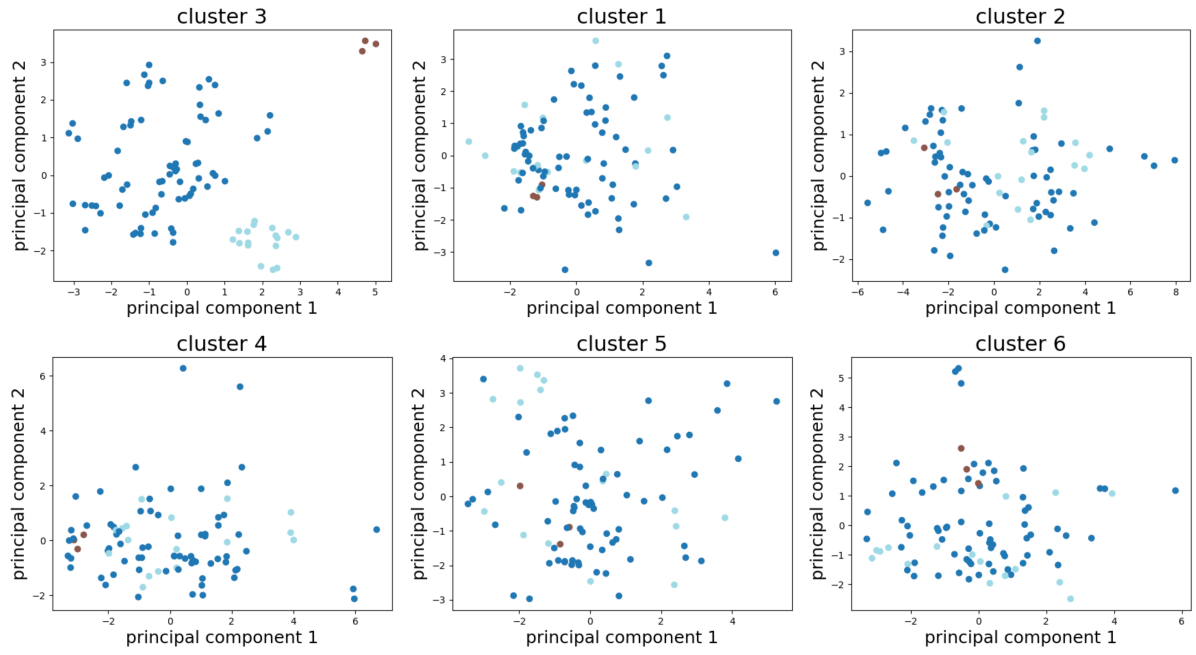


Figure 24: Principal components of F8WTs when grouped into 3 groups, with colors according to groups in c_3